Lecture notes. Lecture 9

Reservoir computing with echo state networks, cellular automata and qubits

Marcin Płodzień
Deep Learning in Classical and Quantum Physics
Institute of Theoretical Physics, Jagiellonian University in Kraków

1 The cost of trained memory

A forecast of a measured record needs the recent history of the record, and the standard tool is a recurrent neural network (RNN). The network carries the history in a state that it updates at every step [5],

$$\mathbf{h}_k = \tanh\left(W_h \mathbf{h}_{k-1} + \mathbf{w}_x s_k + \mathbf{b}_h\right), \qquad \tilde{\mathbf{y}}_k = W_y \mathbf{h}_k + \mathbf{b}_y, \tag{1.1}$$

where s_k is the value of the record at step k, $\mathbf{h}_k$ the hidden state and $\tilde{\mathbf{y}}_k$ the forecast of the next values. All weights are fitted by gradient descent through the whole history, and the gradient with respect to an old input is a product of one derivative per step. This product shrinks or grows exponentially with the age of the input, the problem of vanishing and exploding gradients [6, 7]. Gated networks, the long short-term memory network (LSTM) [8] and the gated recurrent unit (GRU) [9], ease this problem with units whose gates decide what is stored and what is passed on. Every weight of the recurrence and of the gates must still be trained and its hyperparameters optimised. Each pass of gradient descent runs through the whole history, which makes the training slow and expensive.

1.1 Linear evolution of observables

A different view of forecasting avoids the recurrence altogether. In Ref. [11] it was shown that the evolution of a dynamical system acts linearly on functions of its state. For any deterministic map $\mathbf{v} \mapsto F(\mathbf{v})$ a function g of the state, an observable, becomes $g \circ F$ one step later. The Koopman operator $\mathcal{K}g := g \circ F$ is linear in g, however nonlinear F is. When a finite set of observables spans a subspace that $\mathcal{K}$ maps into itself, an invariant subspace, the dynamics restricted to these observables is a matrix. A forecast is then a linear map.

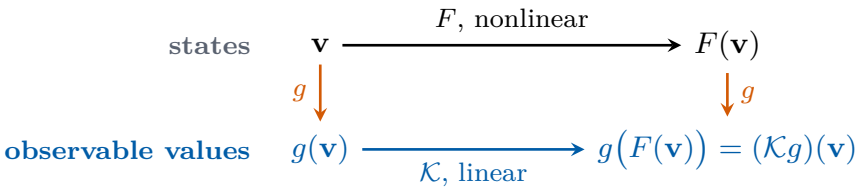


Figure 1: The Koopman operator. The map F moves the state $\mathbf{v}$ nonlinearly (top). An observable g maps every state to a number, and the values of all observables move by the linear operator $\mathcal{K}$, $(\mathcal{K}g)(\mathbf{v}) = g(F(\mathbf{v}))$ (bottom).

As an example, let two numbers evolve by

$$A_{k+1} = \gamma_1 A_k, \qquad Z_{k+1} = \gamma_2 Z_k + A_k^2, \tag{1.2}$$

with constants γ_1 and γ_2. The map is nonlinear because of the square. The three observables

(A, Z, A^2), however, evolve by a matrix,

$$\begin{pmatrix} A \\ Z \\ A^2 \end{pmatrix}_{k+1} = \begin{pmatrix} \gamma_1 & 0 & 0 \\ 0 & \gamma_2 & 1 \\ 0 & 0 & \gamma_1^2 \end{pmatrix} \begin{pmatrix} A \\ Z \\ A^2 \end{pmatrix}_{k}, \tag{1.3}$$

since $A^2_{k+1} = \gamma_1^2 A_k^2$. Applying the matrix τ times gives $Z_{k+\tau}$ as a fixed linear combination of the three numbers at step k, for every horizon. The extra observable A^2 lifts the system into three dimensions, where its evolution is linear.

For irregular motions a finite set of observables that closes in this way is in general not available, since the space of all observables has infinitely many dimensions. A finite set on which the evolution acts approximately linearly still turns forecasting into an approximate linear regression. The linear autoregressive model (AR) uses the last values of the record as such observables, and nonlinear vector autoregression (NVAR) adds their products [10]. Both choose the observables in advance.

1.2 Reservoir computing

A reservoir lets a fixed dynamical system make the observables. The record drives a system whose law is drawn at random once and never trained. Its state, a point in a space of many dimensions, is a set of nonlinear functions of the input history (Figure 2). The echo state network of Figure 3 is the standard example. We fit only the linear readout from this state to the forecast, which takes one linear solve, and no gradient passes through the untrained dynamics. A good reservoir is a driven system whose state is a reproducible and varied nonlinear function of the recent input history.

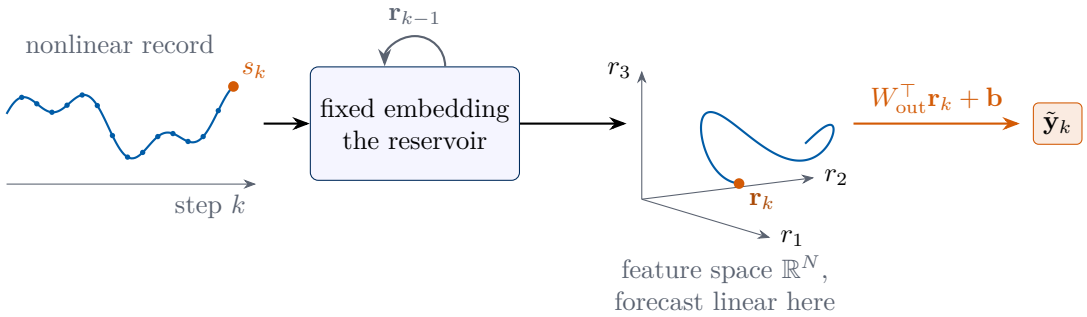


Figure 2: Reservoir computing. A fixed, untrained system embeds the input history into a state $\mathbf{r}_k$ in a space of N dimensions, drawn here for three components. Only the linear readout W_{out} (orange) is fitted.

2 A fixed random system as a feature map

2.1 Features and a linear readout

Every forecasting model here has two parts. A feature map turns the past of the record into a vector $\mathbf{x}_k \in \mathbb{R}^M$ of M numbers, the features, and a linear readout turns the features into forecasts,

$$\tilde{\mathbf{y}}_k = W_{\text{out}}^\top \mathbf{x}_k + \mathbf{b}, \tag{2.1}$$

with an $M \times 50$ matrix of readout weights W_{out} and an offset $\mathbf{b} \in \mathbb{R}^{50}$. The forecast vector $\tilde{\mathbf{y}}_k = (\tilde{s}_{k+1}, \ldots, \tilde{s}_{k+50})$ predicts the target $\mathbf{y}_k = (s_{k+1}, \ldots, s_{k+50})$ from $s_k, s_{k-1}, \ldots$ alone. The same five records serve all models. Lorenz is the coordinate x of the Lorenz-63 model [3], $\dot{x} = 10(y - x)$, $\dot{y} = x(28 - z) - y$, $\dot{z} = xy - \frac{8}{3}z$. This three-variable model of convection has a trajectory that circles one of two centres and jumps irregularly to the other. We integrate it numerically and sample it every $\Delta t = 0.02$. MG 15, MG 25 and MG 35 come from the Mackey–Glass equation [4], $\dot{s}(t) = 0.2\, s(t-\tau_{\text{MG}})/[1 + s(t-\tau_{\text{MG}})^{10}] - 0.1\, s(t)$, a model of blood-cell production whose rate depends

on the value a delay τ_{MG} earlier. We sample it once per time unit at the delays $\tau_{\mathrm{MG}} = 15$, 25 and 35. At $\tau_{\mathrm{MG}} = 15$ the signal repeats itself, and at 25 and 35 it is irregular. Santa Fe is the measured intensity of a far-infrared ammonia laser [1], regular pulsations whose amplitude grows until it collapses at irregular times. Every record has 4000 samples, split in time order. The first 3000 form the training block, whose last 500 serve as the validation block, and the last 1000 form the test block. Each record is rescaled to $[0, 1]$ with the minimum and maximum of the training block, so test values may lie slightly outside this interval. A forecast is made at an origin k, the last step whose value the model may use, and the origins are placed so that no target crosses a block boundary. The fit uses the origins $k = 63, \ldots, 2449$, the validation the origins $2499, \ldots, 2949$ and the test the origins $2999, \ldots, 3949$, the same for all models. The score at the horizon τ is the capacity

$$C(\tau) \coloneqq \operatorname{corr}\left(\tilde{s}_{k+\tau}, s_{k+\tau}\right)^2, \qquad C_{\mathrm{tot}} \coloneqq \sum_{\tau=1}^{50} C(\tau), \tag{2.2}$$

the squared Pearson correlation between forecast and truth over the test origins, so that C_{tot} lies between 0 and 50. Persistence, the forecast that repeats the present value, is the level a useful model must clearly exceed. AR of order p uses $\mathbf{x}_k = (s_k, \ldots, s_{k-p+1})$, and NVAR adds all products $s_{k-i}s_{k-j}$. A reservoir takes as features the state of a driven dynamical system,

$$\mathbf{r}_k = \mathcal{F}(\mathbf{r}_{k-1}, s_k), \qquad \mathbf{x}_k = \mathbf{r}_k, \tag{2.3}$$

whose law $\mathcal{F}$ is drawn at random once. Each component $r_{k,i}$ is an observable of the state of the driven system. Because this state is a function of the input history, the component is also an observable of the history in the sense of Koopman. The reservoir state supplies the lifted coordinates for the linear readout, which forecasts well when the observables that a forecast needs lie close to the span of the reservoir states. Nothing guarantees that a random set of observables contains the right ones [12], so the quality of a reservoir has to be measured.

2.2 Fitting the readout by ridge regression

The readout is fitted by least squares with a penalty on the size of the weights, a method called ridge regression [13]. The matrix X has one row for every training origin, the feature vector $\mathbf{x}_k^\top$, and the matrix Y has the target vectors $\mathbf{y}_k^\top$ as its rows. The weights minimise

$$E(W_{\mathrm{out}}, \mathbf{b}) = \sum_k \left\| W_{\mathrm{out}}^\top \mathbf{x}_k + \mathbf{b} - \mathbf{y}_k \right\|^2 + \beta \left\| W_{\mathrm{out}} \right\|_F^2, \tag{2.4}$$

where $\|W_{\mathrm{out}}\|_F^2$, the squared Frobenius norm, is the sum of the squares of all weights and $\beta > 0$ is the ridge strength. Setting the derivative with respect to the offset to zero gives $\sum_k (W_{\mathrm{out}}^\top \mathbf{x}_k + \mathbf{b} - \mathbf{y}_k) = 0$, so

$$\mathbf{b} = \bar{\mathbf{y}} - W_{\mathrm{out}}^\top \bar{\mathbf{x}}, \tag{2.5}$$

with the means $\bar{\mathbf{x}}$ and $\bar{\mathbf{y}}$ over the training origins. Substituting this offset back replaces every feature vector and every target by its deviation from the mean. With the centred matrices X_c and Y_c the error becomes $\|X_c W_{\mathrm{out}} - Y_c\|_F^2 + \beta \|W_{\mathrm{out}}\|_F^2$, and its derivative with respect to W_{out}, $2X_c^\top (X_c W_{\mathrm{out}} - Y_c) + 2\beta W_{\mathrm{out}}$, vanishes for

$$\left(X_c^\top X_c + \beta\, \mathbb{1}\right) W_{\mathrm{out}} = X_c^\top Y_c, \tag{2.6}$$

where $\mathbb{1}$ is the $M \times M$ identity matrix. Equation (2.6) is a set of linear equations, the normal equations, and the fit is one linear solve. All 50 horizons share the matrix on the left and differ only in the column of Y_c, so they are solved together.

The centring keeps the offset out of the penalty. The penalty is needed because features of a reservoir are often nearly linear combinations of each other. The matrix $X_c^\top X_c$ then has eigenvalues close to zero. Without the penalty the solution would have large weights of opposite signs that cancel on the training data and amplify every small change of the features. Adding $\beta\mathbb{1}$ raises every eigenvalue by β. For one feature the solution reads $w = \sum_k x_{c,k} y_{c,k} / (\sum_k x_{c,k}^2 + \beta)$, the least-squares slope shrunk towards zero by the penalty. We choose the ridge strength on the validation block.

3 The echo state network

A forecast of the laser needs inputs of several ages weighted differently and products of past values. The echo state network supplies both with many units of nonlinear response, coupled to each other at random.

3.1 Definition of the echo state network

The echo state network (ESN) [14, 15] couples N leaky units through a hyperbolic tangent and a fixed random matrix.

Echo state network

The state $\mathbf{r}_k \in \mathbb{R}^N$ is driven by the record through

$$\mathbf{r}_k = (1-\alpha)\,\mathbf{r}_{k-1} + \alpha \tanh\left(W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k\right), \qquad W := \rho\,\widetilde{W}. \tag{3.1}$$

$\widetilde{W}$ is a sparse random $N \times N$ matrix with 10% nonzero entries drawn from a Gaussian distribution, divided by the largest modulus of its eigenvalues. Its eigenvalues are written μ_i, so that $\max_i |\mu_i| = 1$. The matrix W then has the eigenvalues $\rho\mu_i$, and $\rho = \max_i |\rho\mu_i|$ is its spectral radius, the largest modulus of its eigenvalues. The input weights $\mathbf{w}_{\text{in}} \in \mathbb{R}^N$ are drawn uniformly from $[-\sigma_{\text{in}}, \sigma_{\text{in}}]$, the input scaling. With the leak rate $\alpha \in (0, 1]$ each unit keeps the fraction $1-\alpha$ of its old state. W and $\mathbf{w}_{\text{in}}$ are drawn once and never trained, and only the readout of equation (2.1) with $\mathbf{x}_k = \mathbf{r}_k$ is fitted by ridge regression.

The record flows into every unit with its own weight. Every unit flows into the others through the matrix W, whose entry W_{ij} says how strongly and with which sign unit j feeds unit i. The hyperbolic tangent limits this inflow however strongly it is pushed (Figure 3). For $\alpha = 1$ the update is the recurrent network of equation (1.1) without the offset, with random weights that are never changed.

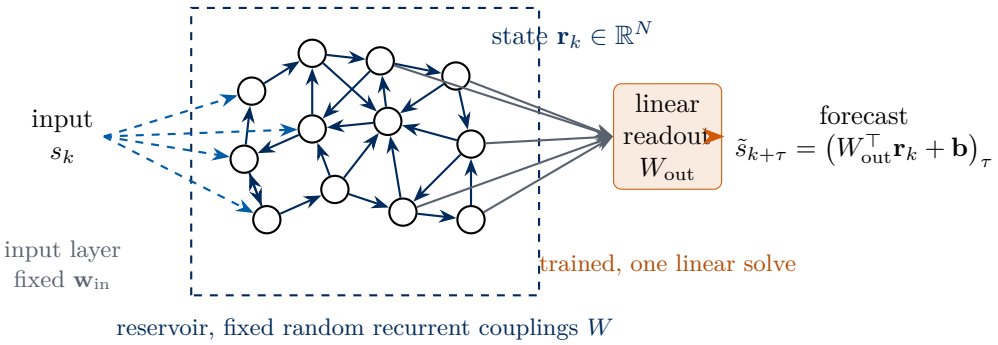


Figure 3: An echo state network. The input s_k enters through fixed weights $\mathbf{w}_{\text{in}}$ (dashed), and the reservoir with fixed random couplings W turns the input history into the nonlinear state $\mathbf{r}_k \in \mathbb{R}^N$. The linear readout W_{out}, the only trained part, maps this state to the forecasts $\tilde{s}_{k+\tau}$.

3.2 Leak rate and spectral radius

The leak rate α says how fast each unit follows its drive. Component i of equation (3.1) reads

$$r_{k,i} = (1-\alpha)\,r_{k-1,i} + \alpha\, g_{k,i}, \qquad g_{k,i} := \left[\tanh\left(W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k\right)\right]_i, \qquad 0 < \alpha \le 1, \tag{3.2}$$

with the state $r_{k,i}$ of unit i at step k and the new drive $g_{k,i}$ that this unit receives from the input and from the other units. The unit keeps the fraction $1-\alpha$ of its old value and adds the fraction α of the new drive. For a given sequence of drives, inserting the equation into itself k times gives

$$r_{k,i} = (1-\alpha)^k\, r_{0,i} + \alpha \sum_{j=0}^{k-1} (1-\alpha)^j\, g_{k-j,i}. \tag{3.3}$$

The start $r_{0,i}$ enters with the factor $(1-\alpha)^k$ and is forgotten exponentially in k for $\alpha < 1$, and after one step for $\alpha = 1$. A drive j steps old enters with the weight $\alpha(1-\alpha)^j$. Since these weights sum to one, $\alpha \sum_{j\geq 0}(1-\alpha)^j = 1$, a unit started long ago is an exponentially weighted average of its past drives, in which recent drives count most. The weights fall by the factor e over

$$-\frac{1}{\ln(1-\alpha)} \approx \frac{1}{\alpha} \qquad (\alpha \ll 1) \tag{3.4}$$

steps, the memory time of the unit, which is 9.5 steps for $\alpha = 0.1$ and 2.8 steps for $\alpha = 0.3$.

In Figure 4 a unit at rest receives one drive $g_{10,i} = 1$ at the step $k = 10$ and no drive at any other step. By equation (3.3) its state is zero before $k = 10$ and $\alpha(1-\alpha)^{k-10}$ afterwards, and its impulse response equals the weight of an input of age $j = k - 10$. For $\alpha = 1$ the unit copies the present drive and has no memory of its own. For $\alpha = 0.1$ the response starts lower and lasts about ten steps.

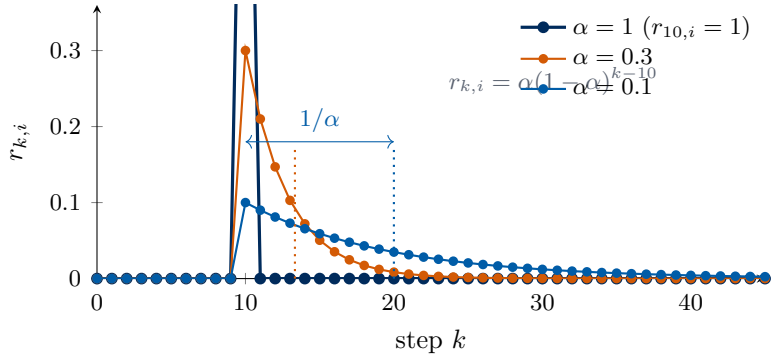


Figure 4: Impulse response of one leaky unit, equation (3.2), started at $r_{0,i} = 0$ and driven by $g_{k,i} = 1$ at $k = 10$ and $g_{k,i} = 0$ at all other steps. The state $r_{k,i}$ against the step k for $\alpha = 1$, 0.3 and 0.1, computed from $r_{k,i} = \alpha(1-\alpha)^{k-10}$ for $k \geq 10$. For $\alpha = 1$ the state is 1 at $k = 10$ only, and the vertical axis cuts this spike. Dotted, the steps $k = 10 + 1/\alpha$, one memory time after the input.

As an analogy, equation (3.2) is a resistor–capacitor circuit, $\tau \dot{V} = g - V$, integrated with the Euler step Δt, with $\alpha = \Delta t/\tau$ the ratio of the time step to the relaxation time. A small α acts as a large capacitor, which keeps a pulse for many steps and smooths a noisy drive, and $\alpha = 1$ removes the capacitor.

The leak rate fixes the time scale of one unit. The spectral radius ρ says how strongly the units feed back on each other through W, and with it how long the echo of an input circulates among the units. For $\rho < 1$ the feedback alone shrinks every pattern of activity, and the network forgets its past at a rate set by ρ and α together. Near $\rho \approx 1$ the slowest patterns decay over many steps and the memory is long. For $\rho > 1$ the feedback amplifies some patterns, and the state may stop being a function of the input alone. Together, α and ρ set the memory window of the reservoir, the span of past inputs that the state still holds.

3.3 Input scaling and the linear regime

The input scaling σ_{in} sets how far the units are pushed into the curved part of the hyperbolic tangent. For a small σ_{in} the units stay near zero, where $\tanh a \approx a$, and respond almost linearly. For a large one they saturate, and products of inputs appear in their response.

How slowly the reservoir forgets follows from this linear regime. Near the state $\mathbf{r} = 0$ and without input the hyperbolic tangent is close to the identity, and equation (3.1) becomes linear,

$$\mathbf{r}_k \approx \left[(1-\alpha)\,\mathbb{1} + \alpha\rho\,\widetilde{W}\right]\mathbf{r}_{k-1}. \tag{3.5}$$

An eigenvector of $\widetilde{W}$ with eigenvalue μ_i is also an eigenvector of the matrix in brackets, with the eigenvalue

$$(1-\alpha) + \alpha\rho\,\mu_i. \tag{3.6}$$

A small perturbation along this eigenvector is multiplied by this number at every step. When the eigenvalue of largest modulus is real and positive, $\mu_i = 1$, the slowest perturbation shrinks by the factor $(1-\alpha) + \alpha\rho$ per step. For $\alpha < 1$ and $\rho < 1$ this factor lies between ρ and 1, so a small leak rate lengthens the memory. The number of steps over which a perturbation falls by the factor e is the memory time $-1/\ln[(1-\alpha) + \alpha\rho]$.

A large random matrix has its eigenvalues spread over a disc in the complex plane (Figure 5), and the eigenvalue of largest modulus is in general complex. For a complex μ_i and $\alpha < 1$ the modulus of the number (3.6) is smaller than $(1-\alpha) + \alpha\rho$, and the formula for the memory time becomes an upper bound. A small leak rate still lengthens the memory.

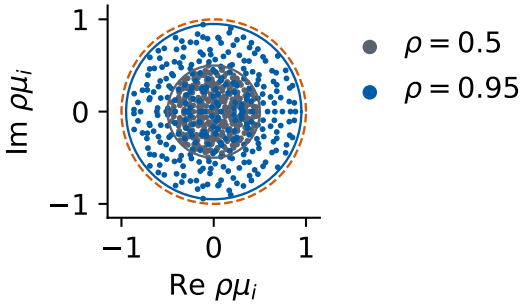


Figure 5: Eigenvalues $\rho\mu_i$ of W for $N = 300$ units in the complex plane, for $\rho = 0.5$ (grey) and 0.95 (blue), with circles of radius ρ and the unit circle (dashed orange).

3.4 Two units

For a reservoir of two units, with values chosen for illustration, we take

$$W = \begin{pmatrix} 0.3 & 1.8 \\ 0.3 & 0 \end{pmatrix}, \qquad \mathbf{w}_{\text{in}} = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix}. \tag{3.7}$$

The eigenvalues $\rho\mu_i$ of W solve $(\rho\mu)^2 - 0.3\,\rho\mu - 0.54 = 0$, which gives $\rho\mu_1 = 0.9$ and $\rho\mu_2 = -0.6$, so the spectral radius is $\rho = 0.9$. For $\alpha = 1$ the two modes decay by the factors 0.9 and -0.6 per step, the second with a change of sign at every step. For $\alpha = 0.5$ the factors (3.6) become 0.95 and 0.2. The leak doubles the memory time of the slow mode and removes the sign changes of the fast one.

3.5 Response to a record and the washout

Figure 6 shows a reservoir started from $\mathbf{r}_0 = 0$ and driven by the laser record. Every unit follows the pulsations with its own amplitude, sign and phase, so the readout receives many different nonlinear functions of the record. The smaller leak rate smooths and slows the response. The first tens of steps depend visibly on the arbitrary start $\mathbf{r}_0 = 0$, and these states, the washout, are left out of the fit. In the forecasts of the five records the washout is the first 63 states, the steps before the first fitting origin.

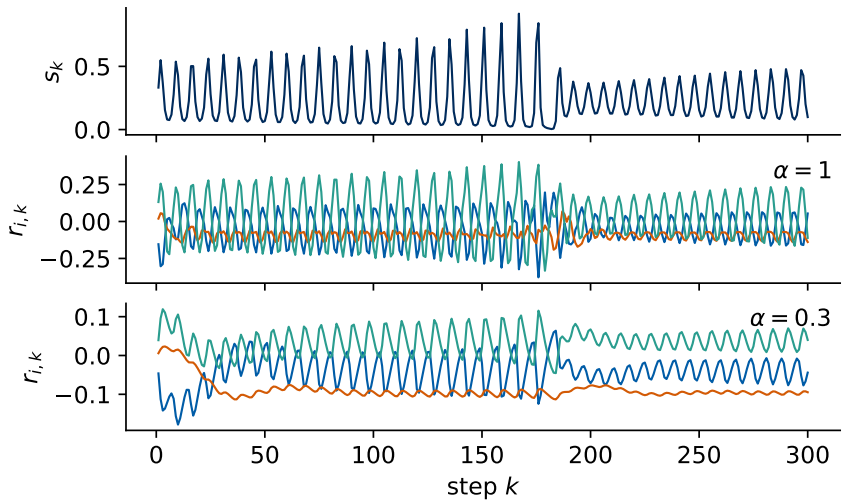


Figure 6: Top, the laser record s_k against the step k. Middle and bottom, three components $r_{k,i}$ of the state of an echo state network for $\alpha = 1$ and $\alpha = 0.3$ ($N = 300$, $\rho = 0.95$, $\sigma_{\text{in}} = 0.5$).

4 The echo state property

The washout of Figure 6 rests on an assumption, that the influence of the arbitrary start $\mathbf{r}_0 = 0$ dies out after some steps. Nothing in equation (3.1) guarantees it.

4.1 Two copies of one reservoir

Two copies of one reservoir, sharing W and $\mathbf{w}_{\text{in}}$, start from two different states $\mathbf{r}_0$ and $\mathbf{r}'_0$ and receive the same record. If the distance between their states shrinks to zero, the record alone fixes the state at late times (Figure 7). The state is then an echo of the input history, hence the name of the network.

Echo state property

Let the admissible input sequences be the sequences with values in a bounded interval. A reservoir has the echo state property when, for every admissible sequence and every pair of starting states, the two copies approach each other,

$$\|\mathbf{r}_k - \mathbf{r}'_k\| \to 0 \qquad (k \to \infty). \tag{4.1}$$

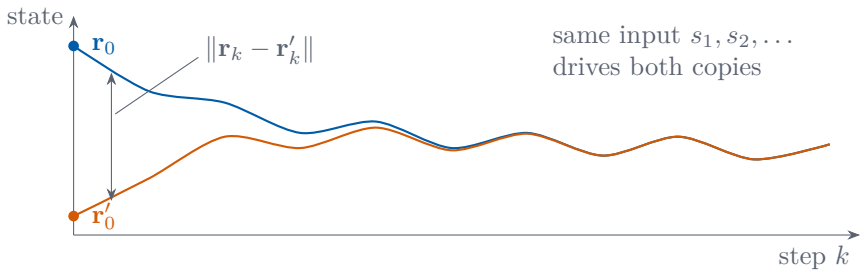


Figure 7: The echo state property, sketched for one component of the state against the step. Two copies of one reservoir start from $\mathbf{r}_0$ and $\mathbf{r}'_0$ and receive the same input.

A readout fitted on one run of the reservoir has to work on another run that starts from another state or another point of the record. Such a transfer requires a state that does not depend on the start, and the washout rests on this property too.

4.2 A sufficient condition

For $\alpha = 1$ the two copies obey $\mathbf{r}_k = \tanh(W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k)$ and $\mathbf{r}'_k = \tanh(W\mathbf{r}'_{k-1} + \mathbf{w}_{\text{in}} s_k)$. By the mean value theorem $\tanh a - \tanh b = \tanh'(c)\,(a - b)$ for some c between a and b, and since $0 < \tanh' \leq 1$,

$$|\tanh a - \tanh b| \leq |a - b|. \tag{4.2}$$

Applied to every component, with $\mathbf{a} = W\mathbf{r}_{k-1} + \mathbf{w}_{\text{in}} s_k$ and $\mathbf{a}' = W\mathbf{r}'_{k-1} + \mathbf{w}_{\text{in}} s_k$, this gives $\|\mathbf{r}_k - \mathbf{r}'_k\| \leq \|\mathbf{a} - \mathbf{a}'\|$. The input is the same in both copies and drops out of the difference, $\mathbf{a} - \mathbf{a}' = W(\mathbf{r}_{k-1} - \mathbf{r}'_{k-1})$. The largest singular value $\sigma_{\max}(W)$ is the largest factor by which W stretches a vector. Hence

$$\|\mathbf{r}_k - \mathbf{r}'_k\| \leq \sigma_{\max}(W)\,\|\mathbf{r}_{k-1} - \mathbf{r}'_{k-1}\| \leq \cdots \leq \sigma_{\max}(W)^k\,\|\mathbf{r}_0 - \mathbf{r}'_0\|, \tag{4.3}$$

which goes to zero for every input when $\sigma_{\max}(W) < 1$. For the leaky update the difference of the two copies is $(1 - \alpha)(\mathbf{r}_{k-1} - \mathbf{r}'_{k-1})$ plus α times the difference of the hyperbolic tangents. Repeating the argument gives the factor $(1 - \alpha) + \alpha\,\sigma_{\max}(W)$ per step, which is below 1 under the same condition and approaches 1 as $\alpha \to 0$. A small leak rate slows the contraction.

The condition is sufficient and often far from necessary. The largest singular value is at least the spectral radius, and for a matrix far from symmetric it can be much larger. The two-unit reservoir of equation (3.7) has $\rho = 0.9$ but $\sigma_{\max} \approx 1.83$, and the bound (4.3) says nothing for it. One step without input can move two of its states further apart. In the linear regime without input, near $\mathbf{r} = 0$, the two copies still converge over many steps, because the k-th power of W shrinks every difference by about ρ^k. With an input the nonlinear update replaces W by a product of different factors at every step, and this estimate no longer applies. The convergence has to be tested.

4.3 A necessary condition

At zero input, $s_k = 0$ for all k, the state $\mathbf{r} = 0$ is a fixed point of equation (3.1), since $\tanh 0 = 0$. Near it the update is the linear map (3.5), whose eigenvalues are the numbers (3.6). When one of them has a modulus above 1, for instance when $\rho > 1$ and the eigenvalue of largest modulus is real and positive, the fixed point is unstable. A copy started at $\mathbf{r}_0 = 0$ then stays at zero forever, while a copy started next to it moves away. The two-unit reservoir rescaled to $\rho = 1.2$ shows this. At zero input a copy started slightly off zero settles at a fixed point away from zero, and a copy started on the opposite side settles at its mirror image. A copy started at zero stays there. Whenever the zero sequence is an admissible input, $\rho \leq 1$ is therefore necessary when the leading eigenvalue is real and positive.

Between the sufficient condition $\sigma_{\max}(W) < 1$ and the necessary condition at zero input there is a gap, and no condition is known that is both necessary and sufficient [16]. In practice ρ is used as a control parameter, and the property is tested directly with the input at hand.

4.4 Testing the property

Two copies of one reservoir start from random states and are driven by the laser record (Figure 8). For $\rho = 0.8$ they converge to round-off within a few hundred steps and for $\rho = 1$ much more slowly. For $\rho = 1.2$ and 2.0 they never converge, and a readout trained on one run does not transfer to another. The sufficient condition covers only small ρ for this reservoir. The convergence at $\rho = 0.8$ and 1 is a property of this reservoir and input that the bound does not guarantee.

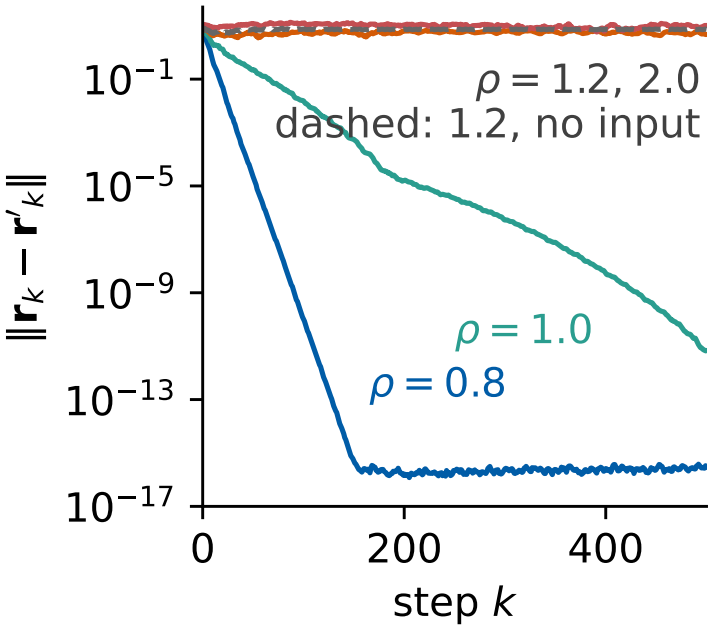


Figure 8: Test of the echo state property. The distance $\|\mathbf{r}_k - \mathbf{r}'_k\|$ between the two copies against the step k, on a logarithmic scale, for four spectral radii. Dashed, $\rho = 1.2$ without input ($N = 100$, $\alpha = 1$, $\sigma_{\text{in}} = 0.5$).

4.5 Contraction under a strong input

Repeated on a grid of spectral radii for four input scalings (Figure 9), the distance drops to round-off below a threshold value of ρ and jumps to order one above it. The threshold moves to larger ρ as the input scaling grows.

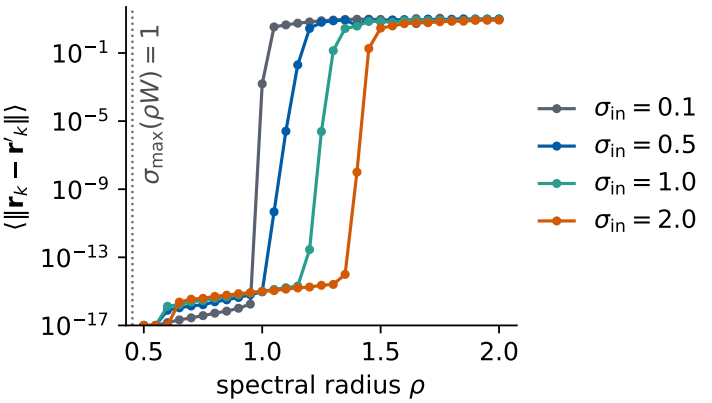


Figure 9: The median over three pairs of starts of the mean distance between the two copies at late times, against ρ, for the reservoir of Figure 8 and four input scalings σ_{in}. Dotted, the sufficient bound $\sigma_{\max}(\rho\widetilde{W}) = 1$.

A strong input pushes the units towards saturation, where the derivative $\tanh'$ is small. Two nearby states then differ after one step by about $\Gamma_k W$ times their difference, with the diagonal matrix $\Gamma_k := \operatorname{diag}(1 - \mathbf{r}_k^2)$ of the derivatives $\tanh' = 1 - \tanh^2$. A small Γ_k lowers the effective gain of the recurrence. Driven by a constant input, the two copies of the two-unit reservoir approach each other faster than at zero input. The common rule $\rho < 1$ is therefore a working rule for weak inputs, and with a strong input a reservoir above $\rho = 1$ can still have the property for that input.

5 Fading memory

A reservoir has to remember the recent past and forget the rest (Figure 10). The value $s_{k+\tau}$ of a chaotic record is not a function of the present value s_k alone, because the same value occurs on different parts of the trajectory. It is a function of a window of recent values $s_k, s_{k-1}, \ldots, s_{k-m}$, the

delay window. By the theorem of Takens, a long enough window of delayed measurements fixes the state of the system that produced the record [18]. Since the linear readout sees only the present state $\mathbf{r}_k$, the state has to carry this window. Everything older, and the start, should be forgotten, for three reasons.

The start $\mathbf{r}_0$ is arbitrary. If it persisted, one input history would lead to different states in different runs, and a readout trained on one run would fail on another. Forgetting the start, the echo state property, makes $\mathbf{r}_k$ a reproducible function of the input history.

Old values of a chaotic record carry little information about its future. Two trajectories of a chaotic system that start a small distance apart separate on average as $e^{\lambda_1 t}$, where $\lambda_1 > 0$ is the largest Lyapunov exponent. A measured value therefore constrains the future for only a few Lyapunov times $T_\lambda \coloneqq 1/\lambda_1$. For Lorenz, $\lambda_1 = 0.904$ and one Lyapunov time is about 55 samples. Old inputs still occupy components of the state, of which a reservoir of N units has only N. Storing the distant past therefore crowds out the recent inputs and their nonlinear combinations.

Feedback that is too strong, $\rho > 1$, amplifies small differences of the state and the noise of the input. The dynamics of the reservoir itself then dominates the input, and the state stops following the signal.

The balance is a memory that fades at a controlled rate. In a leaky unit a drive j steps old weighs $\alpha(1-\alpha)^j$ by equation (3.3), and the spectral radius ρ sets how long echoes survive among the units. The memory window that α and ρ produce should match the window that the forecast needs, about the delay window. It should also stay shorter than the time over which the record decorrelates. As an analogy, a thermometer whose time constant is much longer than the changes it should follow reports a stale average. A thermometer with a much shorter constant reports the noise of the present moment.

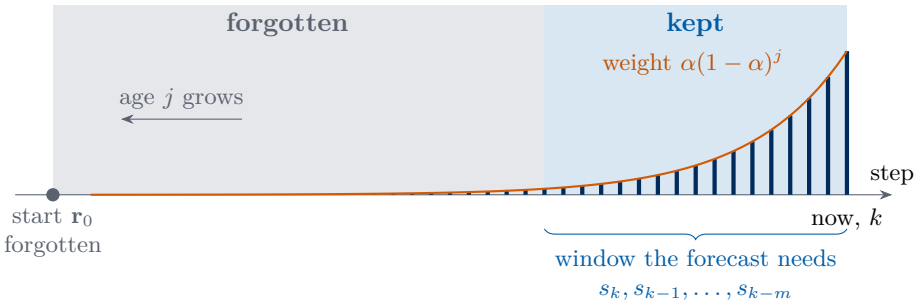


Figure 10: Remembering and forgetting, sketched. The bars are the weights $\alpha(1-\alpha)^j$ with which the inputs enter the present state, against the input step, for an input j steps old. The inputs in the window that the forecast needs (blue) are kept, older inputs (grey) are forgotten, and so is the start $\mathbf{r}_0$.

Any reservoir can be asked to forget old inputs in this way.

5.1 Definition of fading memory

Two records that agree over the recent past and differ only long ago should lead to nearly the same present state (Figure 11).

Fading memory

Let $\mathbf{r}_k(\mathbf{s})$ be the state at step k after the input history $\mathbf{s} = (\ldots, s_{k-1}, s_k)$. The reservoir has a fading memory when histories that agree over the last K steps lead to states whose distance goes to zero as K grows,

$$s_{k-j} = s'_{k-j} \text{ for } 0 \le j < K \quad \Longrightarrow \quad \|\mathbf{r}_k(\mathbf{s}) - \mathbf{r}_k(\mathbf{s}')\| \le \varepsilon_K, \qquad \varepsilon_K \to 0 \ \ (K \to \infty), \tag{5.1}$$

with a bound ε_K that holds for all admissible histories.

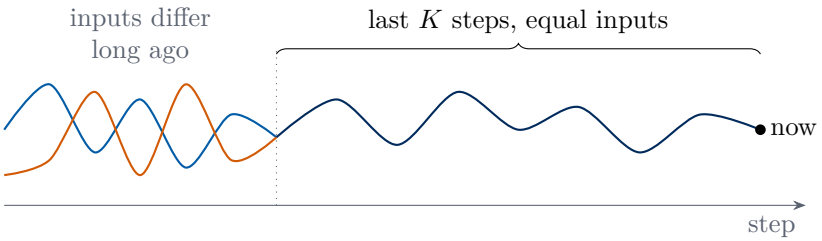


Figure 11: Fading memory, sketched. Two input histories (blue and orange) differ long ago and agree over the last K steps (dark line).

The echo state property compares two starting states under one input, and fading memory compares two input histories that share their recent part. For the leaky unit of equation (3.2) both follow from the sum (3.3). Let the drives be bounded, $|g_{k,i}| \le g_{\max}$, and let two sequences of drives, $\mathbf{g}$ and $\mathbf{g}'$, agree for the last K steps. The two states differ only through the drives of age $j \ge K$, and

$$\begin{aligned} |r_{k,i}(\mathbf{g}) - r_{k,i}(\mathbf{g}')| &\le \alpha \sum_{j \ge K} (1-\alpha)^j \, |g_{k-j,i} - g'_{k-j,i}| \\ &\le 2 g_{\max} \, \alpha \sum_{j \ge K} (1-\alpha)^j = 2 g_{\max} \, (1-\alpha)^K . \end{aligned} \tag{5.2}$$

The bound falls exponentially with K. The argument holds for any reservoir whose update shrinks every difference of states by at least a factor $c < 1$ per step, whatever the common input. Its states lie in a bounded set, since every component of the hyperbolic tangent lies in $[-1, 1]$. At the moment from which two histories agree, their states are therefore at most $2\sqrt{N}$ apart, and K steps of common input shrink this distance to at most $2\sqrt{N}\, c^K$. Such a reservoir has both properties. A reservoir without a uniform contraction, like the reservoirs of Figure 8, has to be tested for each.

5.2 Fading memory and polynomial readouts

Fading memory matters because of a general result [17]. We consider an operator that maps a bounded input history to an output at the present step, does not change its rule in time, and has fading memory. Every such operator can be approximated, uniformly over all bounded inputs and to any required accuracy, by a finite Volterra series,

$$y_k \approx h_0 + \sum_{j=0}^{K-1} h_1(j) \, s_{k-j} + \sum_{j_1, j_2 = 0}^{K-1} h_2(j_1, j_2) \, s_{k-j_1} s_{k-j_2} + \dots , \tag{5.3}$$

a polynomial of finite degree in the inputs of a finite window of K past steps, with coefficients $h_0, h_1, h_2, \dots$ called the kernels. The original result is stated in continuous time, with integrals over the past, and equation (5.3) is its discrete-time analogue. An equivalent form is a bank of linear filters of the input followed by a polynomial readout. A reservoir provides the filters through its memory and the polynomial through its nonlinearity, and a linear readout combines them.

The leaky unit with a quadratic readout, $y_k = r_{k,i}^2$, shows the structure, with its drive $g_{k,i}$ in the role of the input s_k. Inserting the sum (3.3) for a unit started long ago gives

$$y_k = \alpha^2 \sum_{j_1, j_2 \ge 0} (1-\alpha)^{j_1 + j_2} \, g_{k-j_1,i} \, g_{k-j_2,i}, \tag{5.4}$$

a Volterra series with the single kernel $h_2(j_1, j_2) = \alpha^2 (1-\alpha)^{j_1+j_2}$. Cutting the sums at K changes $r_{k,i}$ by at most $g_{\max}(1-\alpha)^K$, the tail of the sum, and since $|r_{k,i}| \le g_{\max}$ it changes $y_k = r_{k,i}^2$ by at most $2 g_{\max}^2 (1-\alpha)^K$. The finite series approximates this operator with an error that falls exponentially with the window, and the window is finite because the memory of the unit fades.

5.3 Testing fading memory

Two copies start from the same state and receive inputs that differ for 50 steps and then coincide (Figure 12). With random inputs the logarithm of the difference of two leaky units falls by $-\ln(1-\alpha)$ per step, as equation (5.2) predicts. An echo state network with $\rho = 0.8$ forgets faster and reaches round-off within about a hundred steps, while at $\rho = 1.2$ the difference fades only slowly (left). With the laser record, whose first values are replaced in one copy, the difference disappears to round-off at $\rho = 0.8$ and persists to the end of the run at $\rho = 1.2$ (right). A spectral radius near or above 1 gives a long memory together with a slow or missing loss of the start and of old inputs.

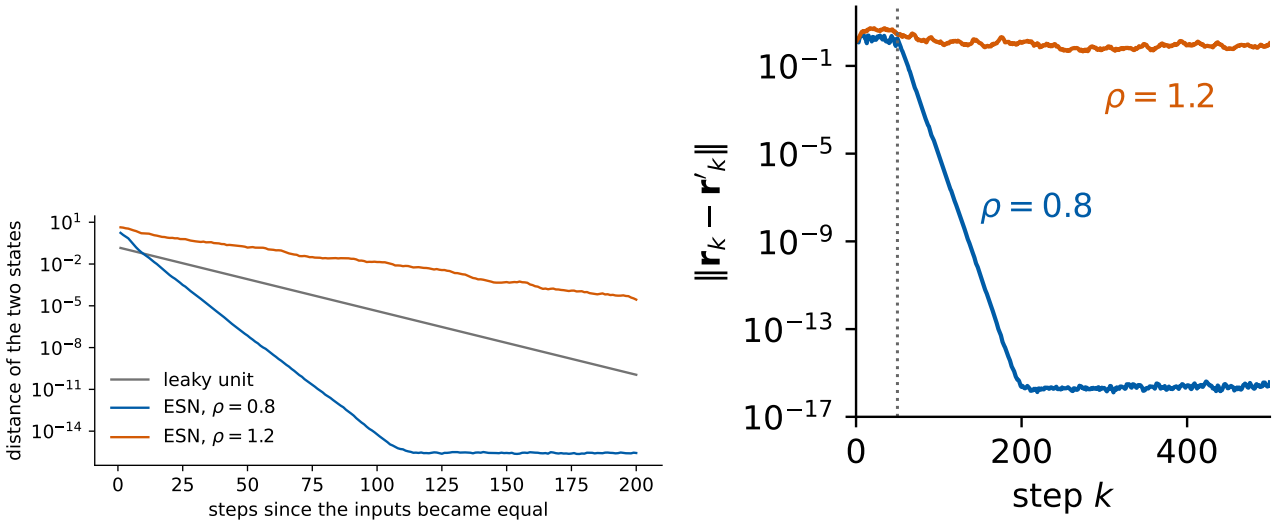


Figure 12: Tests of fading memory. Left, random inputs. The median distance of the two states over 10 pairs of inputs against the number of steps since the inputs became equal. Right, the laser record and the same record with its first 50 values replaced (end of the replacement dotted). The distance $\|\mathbf{r}_k - \mathbf{r}'_k\|$ against the step k.

6 The memory of a reservoir

The window of past inputs that a reservoir still holds is measured with an input that cannot be predicted from its own past. With such an input every recovered past value must come from the state [19]. The input u_k is drawn independently at every step and uniformly from $[-1, 1]$. For every delay d a separate linear readout of the state is fitted to return the past input u_{k-d}, and its quality on test data is the recall capacity

$$C(d) := \operatorname{corr}\left(\tilde{u}_{k-d}, u_{k-d}\right)^2, \qquad \mathrm{MC} := \sum_{d=1}^{100} C(d), \tag{6.1}$$

with the Pearson correlation of equation (2.2) taken over the test steps. The memory capacity MC counts how many past inputs the state holds, where a partly recovered input counts as a fraction. The sum is cut at $d = 100$, far beyond the delays at which $C(d)$ of every reservoir in these notes has dropped to zero.

The sum of $C(d)$ over all delays $d \geq 0$, the present input included, is at most N for a reservoir of N units [14, 19], so $\mathrm{MC} \leq N$ as well. A linear readout combines only N state components, and independent inputs at different delays are independent directions, of which N components cover at most N. A linear reservoir whose matrix shifts the value of every unit to the next reaches the bound. Its state is the delay line $(u_k, u_{k-1}, \ldots, u_{k-N+1})$, which returns the present input and the $N-1$ inputs before it without error. A random matrix of the same size spreads its memory over more delays and recovers less in total (Figure 13).

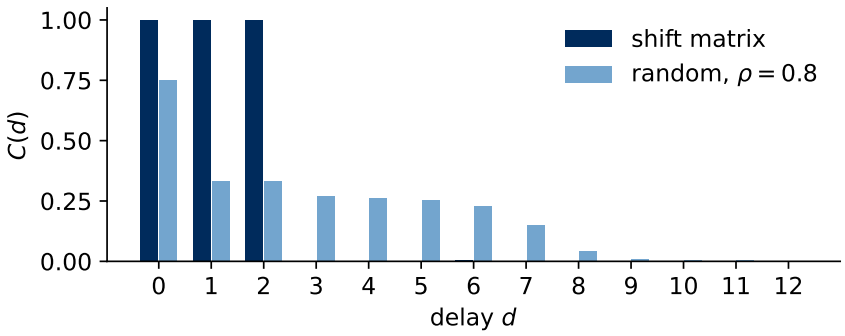


Figure 13: Recall capacity $C(d)$ against the delay d for two linear reservoirs of three units driven by independent uniform inputs, the shift matrix (dark) and a random matrix with $\rho = 0.8$ (light).

For an echo state network the memory curves stay close to 1 for short delays and then drop to zero (Figure 14). The drop moves to longer delays as ρ approaches 1. A longer memory needs a slower relaxation, which a larger spectral radius provides, and the memory capacity stays well below the bound N. Above $\rho = 1$ the state stops being a function of the input alone, and the memory capacity falls and differs widely between random reservoirs. In the left panel the memory capacity grows from 12.3 at $\rho = 0.5$ to 19.9 at $\rho = 0.95$ and drops to 16.8 at $\rho = 1.1$.

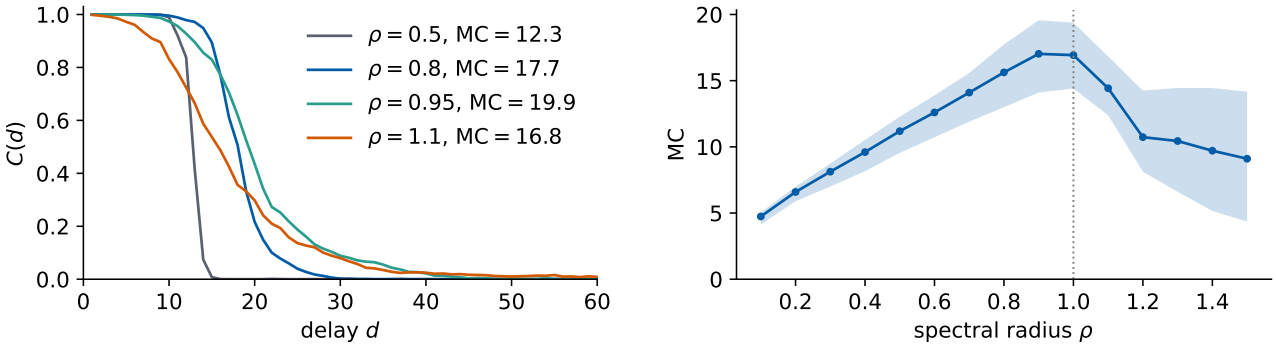


Figure 14: Memory of an echo state network with 50 units, $\alpha = 1$ and $\sigma_{\text{in}} = 0.1$. Left, the recall capacity $C(d)$ against the delay d for four spectral radii, with the memory capacity in the legend. Right, the memory capacity against ρ, mean (line) and range (band) over three random reservoirs. Dotted, $\rho = 1$.

7 Forecasting the five records

We test the reservoir on the forecast of the five records, scored by the capacity of equation (2.2) on the test block. We compare its scores with fixed dictionaries of features and with networks whose weights are all trained.

7.1 Forecasting protocol

One random reservoir of $N = 300$ units, drawn once, serves all five records. The record drives it from its first value on, starting at $\mathbf{r}_0 = 0$, and the 63 states before the first fitting origin $k = 63$ form the washout. One readout maps the state to all 50 horizons. The spectral radius, the input scaling, the leak rate and the ridge strength are chosen on a small grid by the validation C_{tot} for each record. The AR and NVAR models serve as baselines, with their order p and ridge strength chosen in the same way.

7.2 Test scores

Table 1 and Figure 15 compare the fixed random reservoir with the fixed dictionaries of features of AR and NVAR. On Lorenz the echo state network reaches $C_{\text{tot}} = 39.2$, against 15.4 for AR and

NVAR. On the periodic MG 15 all three models come close to the maximum, and on the laser the network and NVAR tie. The gain is largest where the forecast needs a long memory of a nonlinear signal. On Lorenz the delay window of AR and NVAR, $p = 16$ lags selected on the validation block, covers only a fraction of a loop around one of the two centres. The fading memory of the reservoir state has no fixed number of lags and can reach further back than these p lags.

test C_{tot}	Lorenz	MG 15	MG 25	MG 35	Santa Fe
persistence	8.4	19.5	13.3	11.8	5.5
AR	15.4	49.8	33.1	38.0	16.5
NVAR	15.4	50.0	45.5	48.6	34.7
ESN, $N = 300$	39.2	50.0	48.3	49.8	34.7
selected ρ	1.1	1.1	0.95	0.8	0.95
selected σ_{in}	1.0	0.5	1.0	1.0	0.1
selected α	1.0	1.0	0.3	0.3	1.0

Table 1: Total capacity on the test block of persistence, AR, NVAR and the echo state network (maximum 50). The lower rows give the reservoir settings chosen on the validation block from $\rho \in \{0.5, 0.8, 0.95, 1.1\}$, $\sigma_{\mathrm{in}} \in \{0.1, 0.5, 1\}$, $\alpha \in \{0.3, 1\}$ and $\beta \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$.

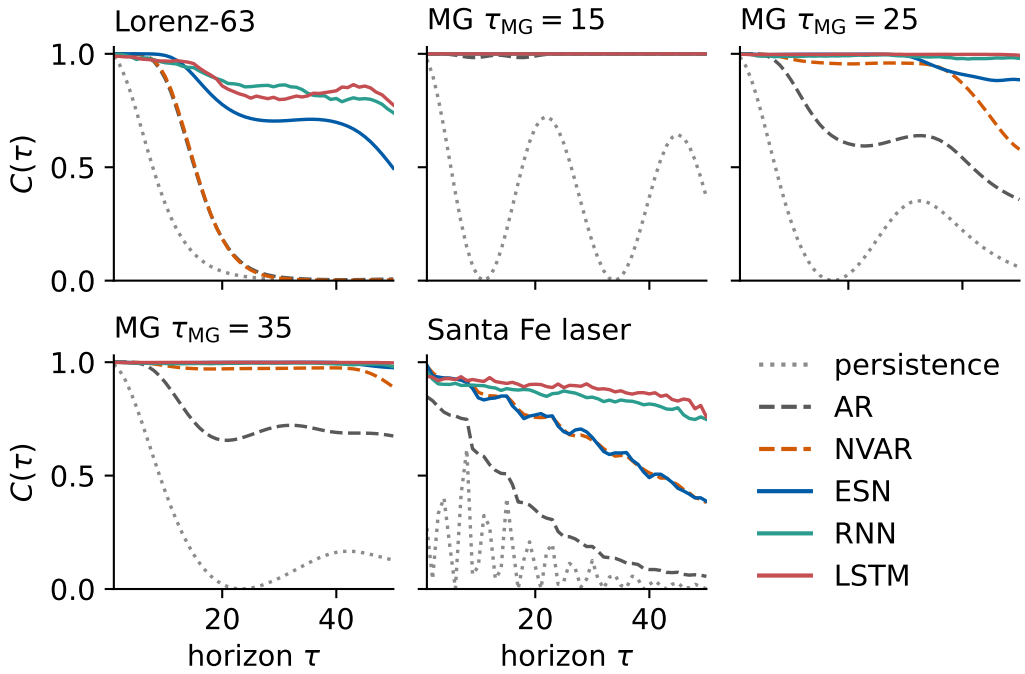


Figure 15: Capacity $C(\tau)$ on the test block against the horizon τ, one panel per record. The curves show persistence, AR, NVAR, the echo state network with 300 units and the trained RNN and LSTM.

The recurrent network of equation (1.1), the GRU and the LSTM have all their weights trained by gradient descent through the history. They were fitted to the same records and scored on the same origins and horizons. At the settings selected for the laser they have between 2738 (RNN) and 6002 (LSTM) trained weights, against 15050 readout weights of the echo state network. They keep a lead on the laser, where the LSTM reaches $C_{\mathrm{tot}} = 44.3$ and the GRU 43.0 against 34.7 for the echo state network. On Lorenz the GRU also leads with 45.1 against 39.2, while on the Mackey–Glass records the echo state network comes close to them. Their training by many passes of gradient descent took many times longer than the one linear solve of the reservoir. The factor depends on the hardware, its load and the implementation.

The selected settings follow the time scales of the records. The slowly varying MG 25 and MG 35

receive the small leak rate, a long memory per unit. The laser receives the weakest input scaling, which keeps the units close to their linear range. Lorenz and MG 15 receive $\rho = 1.1$, above the rule $\rho < 1$, and on both records the two copies of the echo state test agree to round-off on the test origins. At the first training origin the copies still differ, so the first rows of the fit carry a trace of the start. A longer washout changes the test scores by less than one unit.

One step ahead the echo state network forecasts the three Mackey–Glass records almost without error, and their forecasts lie on the diagonal of Figure 16. On the laser the visible scatter comes from the largest pulses and from the collapses.

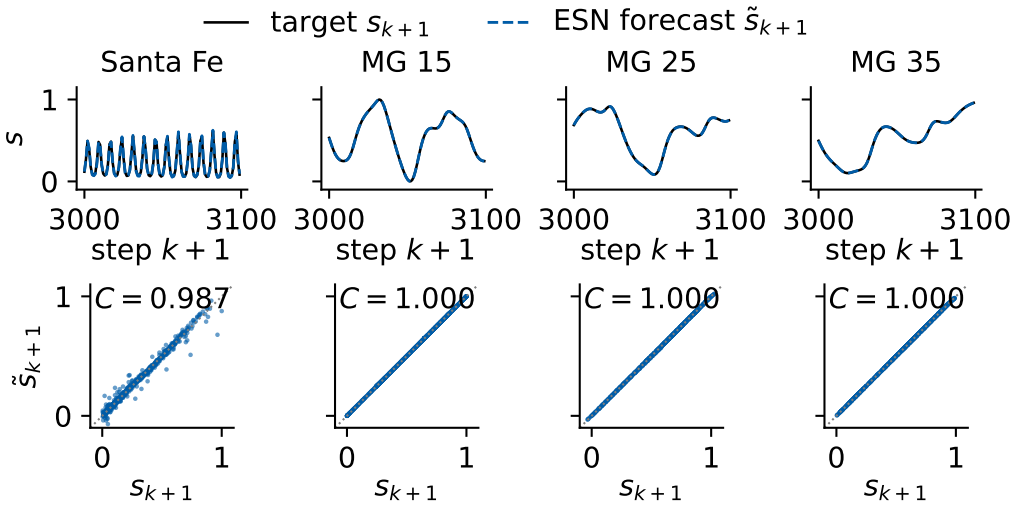


Figure 16: One-step forecasts of the echo state network. Top, the target s_{k+1} (black) and the forecast $\tilde{s}_{k+1}$ (dashed) for the first 100 test targets. Bottom, the forecast against the target on the whole test block, with C at $\tau = 1$.

7.3 Differences between random reservoirs

To measure how much a score depends on the random draw, we draw five reservoirs independently at the same settings. On Lorenz the five scores lie between 39.0 and 40.0, all far above the 15.4 of NVAR. The advantage of the reservoir on this record is therefore independent of the draw. On the laser they lie between 31.5 and 34.7, and the first draw, the one that tied NVAR at 34.7, is the best of the five. The tie belongs to one draw, and on average reservoirs of this size lie below NVAR on the laser. A difference between two models that is smaller than the spread between random reservoirs carries no information about the models.

7.4 Size of the reservoir

Figure 17 shows the validation C_{tot} on a grid of N and ρ, with σ_{in} and α fixed at the values selected for each record. On the laser the best value keeps growing up to the largest reservoir and shows no saturation, while on Lorenz it grows only a little. The laser forecast needs many different nonlinear functions of the last few pulsations, and every added unit contributes such functions. The dependence on the spectral radius follows the echo state property. At the weak input of the laser the scores of the larger reservoirs collapse above $\rho = 1$, where the test of Figure 9 found that the state stops forgetting its start. On Lorenz the strong input keeps good scores up to $\rho = 1.1$.

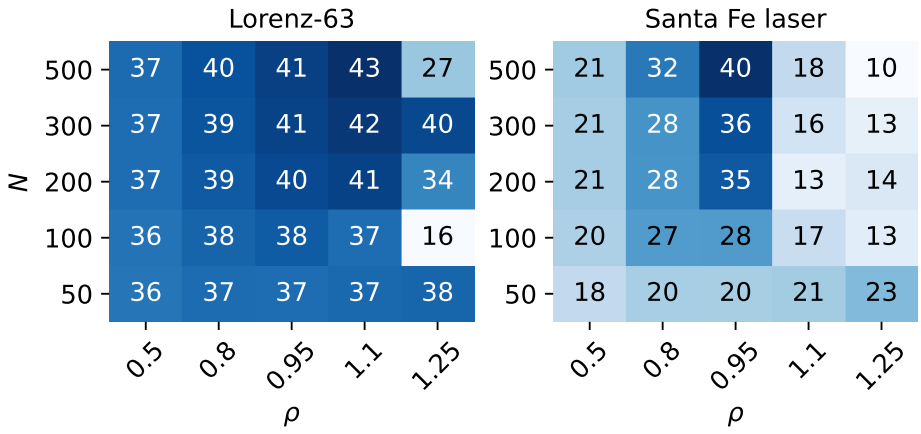


Figure 17: Validation C_{tot} of the echo state network on the grid of N and ρ, for Lorenz (left) and the laser (right).

7.5 Closed loop

In the forecasts above the measured record drove the reservoir, and at every step the readout turned the new state into a forecast. The readout is trained in this mode, the open loop. Once the readout forecasts one step ahead, its output can replace the measurement. In the closed loop the forecast $\tilde{\mathbf{u}}_{k+1}$ is fed back as the next input, and no data enter any more. The reservoir together with its readout then becomes an autonomous dynamical system, $\mathbf{r}_{k+1} = \mathcal{F}(\mathbf{r}_k, W_{\text{out}}^{\top}\mathbf{r}_k + \mathbf{b})$, that generates its own trajectory (Figure 18). The closed loop forecasts beyond the data, since the trained system continues from the last measured state without further measurements. It also turns the reservoir into a model of the system learned from the record alone. This model can be run for a long time to measure properties of the dynamics such as the attractor and the largest Lyapunov exponent [2]. Because every open-loop forecast starts from a measured value, errors never accumulate, and a good one-step forecast does not show that the dynamics was learned. In the closed loop errors are not corrected, and the free run reproduces the attractor only if the trained system has learned the dynamics that produced the record.

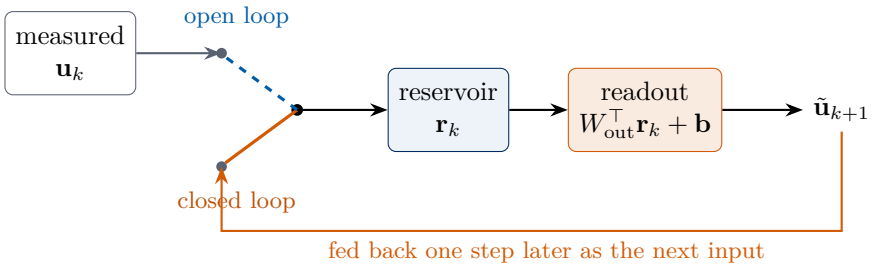


Figure 18: Open and closed loop of a trained reservoir. In the open loop (switch on the upper contact, dashed blue) the measured state $\mathbf{u}_k$ drives the reservoir and the readout is fitted to the target $\mathbf{u}_{k+1}$. In the closed loop (lower contact, orange) the forecast $\tilde{\mathbf{u}}_{k+1}$ is fed back as the next input.

Training and prediction happen in separate phases, and the forecast is fed back only in the second one (Figure 19). In the training phase the measured record $\mathbf{u}_1, \ldots, \mathbf{u}_K$ of the training block drives the reservoir. Its states $\mathbf{r}_k$ are collected after a washout that removes the memory of the arbitrary initial state. The readout is then fitted to the measured next values by minimising

$$L(W_{\text{out}}, \mathbf{b}) = \sum_k \left\| W_{\text{out}}^{\top}\mathbf{r}_k + \mathbf{b} - \mathbf{u}_{k+1} \right\|^2 + \beta \left\| W_{\text{out}} \right\|_F^2, \tag{7.1}$$

the ridge problem of equation (2.6) with the next state as target. Both the states and the targets come from the data, so the forecasts of the reservoir never enter the training. The loss is quadratic in the readout and is minimised once, by one linear solve, after which W_{out} and $\mathbf{b}$ stay fixed. No loss is computed anywhere else.

In the prediction phase the loop is closed. Before each forecast the reservoir is driven by the measured record up to the starting point, the origin, so that its state corresponds to the current state of the system. From the origin on, the forecast $\tilde{\mathbf{u}}_{k+1} = W_{\text{out}}^{\top}\mathbf{r}_k + \mathbf{b}$ is the next input and nothing is fitted. The free run is only compared with the truth, pointwise for the weather and statistically for the climate.

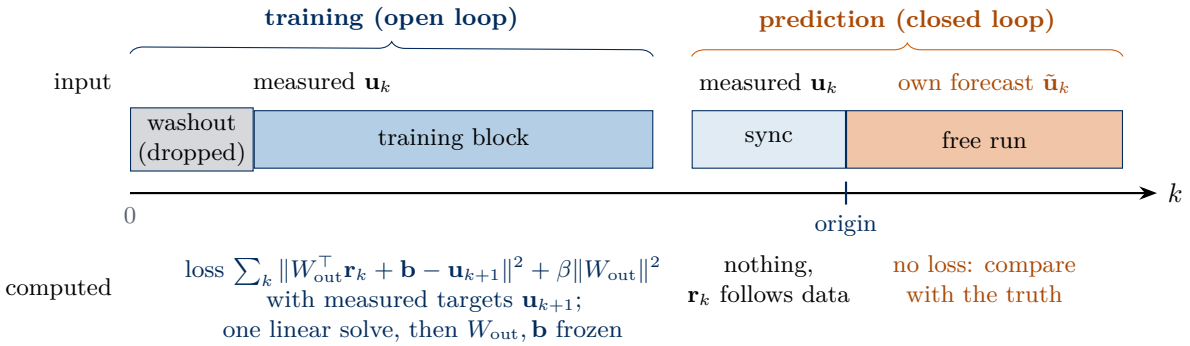


Figure 19: Training and prediction of a closed-loop reservoir along time. Training (open loop), the measured $\mathbf{u}_k$ drives the reservoir, the states of the washout are dropped, and the loss (7.1) with measured targets fixes W_{out} and $\mathbf{b}$ by one linear solve. Prediction (closed loop), measured data synchronise the reservoir up to the origin, after which its own forecast $\tilde{\mathbf{u}}_k$ is the input and the run is compared with the truth.

On Lorenz-63 we train a reservoir of 300 units one step ahead on the three coordinates $\mathbf{u}_k = (x, y, z)$, each normalised by its mean and standard deviation. The trained reservoir is released from a state synchronised with the data. The first test is the weather, how long the free run stays close to the true trajectory started from the same state. No model can follow a chaotic trajectory for ever, because any small error grows like $e^{\lambda_1 t}$ and destroys a pointwise forecast after a few Lyapunov times. The error $e(\tau)$ is the distance between the free run and the truth at the horizon τ, divided by the typical size of the state. The error grows from about 10^{-6} at the start, first quickly and then at close to the rate λ_1 of the Lorenz flow (Figure 20, left). The valid time t_v is the first horizon at which $e(\tau)$ exceeds 0.4. Over 190 starting points its median is $6.9\,T_\lambda$, and 80% of them lie between 4.9 and $8.7\,T_\lambda$ (right). The forecast horizon is set by the chaos of the Lorenz flow, and a smaller initial error would add only a few Lyapunov times.

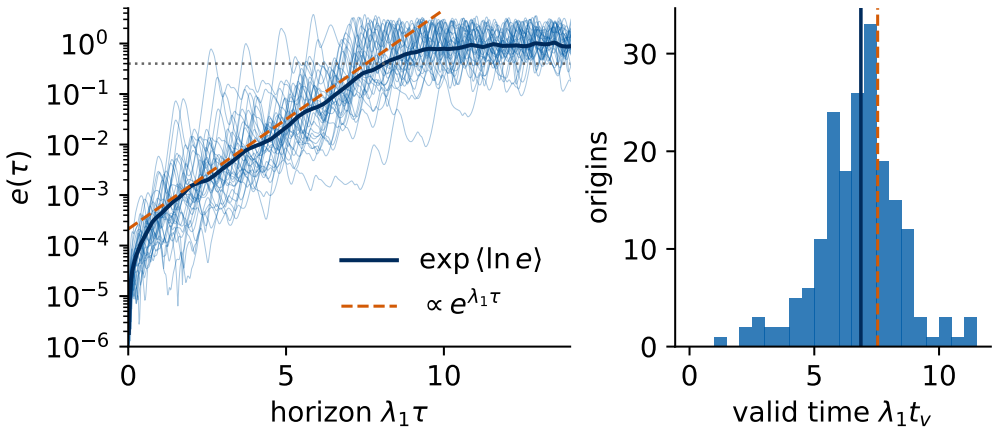


Figure 20: Weather of the closed-loop reservoir on Lorenz-63. Left, the normalised error $e(\tau)$ of the free run against the horizon in Lyapunov times $\lambda_1\tau$. Thin lines show every sixth starting point, the dark line their geometric mean, the dashed line a line of slope λ_1 and the dotted line the threshold 0.4. Right, the histogram of the valid times $\lambda_1 t_v$ over the 190 starting points, with the median (dark line).

For a long run of a chaotic system the meaningful second test is the climate. It asks whether the free run has the statistics of the true attractor after it has lost the trajectory. A free run of about 1988 Lyapunov times fills the two wings around the two centres like a true run (Figure 21). Its histograms of x, y and z agree with the true ones within the histogram resolution. The largest

Lyapunov exponent of the free run, 0.909, agrees with the value 0.904 of the Lorenz equations. The reservoir has learned a dynamical system with the attractor and the rate of chaos of the Lorenz flow. Of 240 reservoirs with different settings and random draws, only 96 ran freely on an attractor like that of Lorenz. The others settled onto a periodic orbit or a fixed point, wandered on a wrong attractor or diverged. A good one-step forecast does not guarantee the right climate, so every free run has to be checked separately.

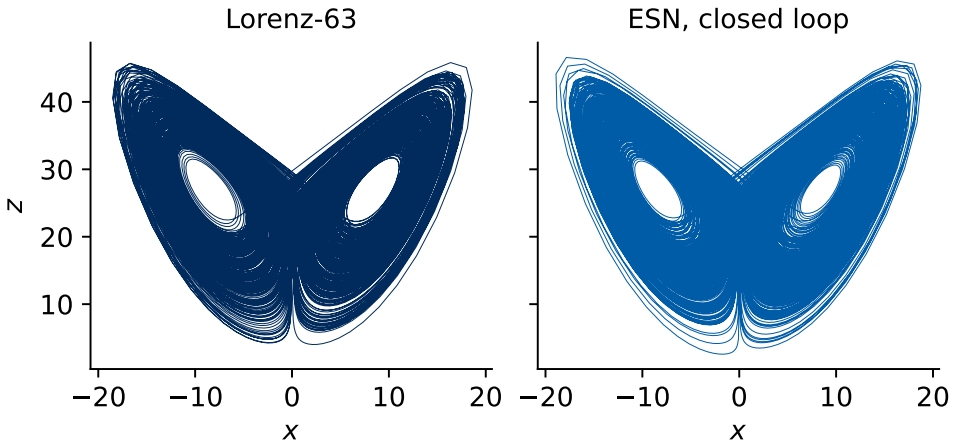


Figure 21: Climate of the closed-loop reservoir. The (x, z) projection of a long run of Lorenz-63 (left) and of the free run of the closed-loop echo state network (right).

8 Requirements and physical reservoirs

The tests of the echo state network set requirements on any driven system that is to serve as a reservoir. The state has to forget its start and its distant input, the echo state property and fading memory. Different recent histories have to lead to different states, since a readout cannot tell apart histories that produce the same state. The response has to be nonlinear, so that products of inputs at different delays appear in the state. The state has to offer many quantities that are not linear combinations of each other, as the growth of the laser score with the number of units showed. Finally the memory time of the system has to match the time scales of the task, the role that the leak rate played on the Mackey–Glass records.

Nothing in these requirements asks for a neural network. In Ref. [21] it was shown that echo state networks with linear readouts are universal approximants of fading-memory operators, and physical systems can take the place of the random network [20]. The surface of water in a tank, driven by motors that encode the input and filmed by a camera whose images serve as features, works as a reservoir [22]. Networks of masses and nonlinear springs driven by the input serve as reservoirs whose positions are read out [24]. A soft silicone arm modelled on an octopus tentacle is moved in water by a motor at its base. It serves as a reservoir when the readings of bending sensors along the arm are read out linearly. Its passive body supplies memory and nonlinearity [23]. A single nonlinear electronic node with a delayed feedback loop also works as a reservoir, the samples along the delay line acting as virtual units [25]. So do memristors, resistors whose resistance depends on the current that has passed through them [26]. In each case only a linear readout of measured quantities is fitted.

9 Cellular automata

The physical reservoirs above are continuous systems, water, springs and soft matter, whose state is a set of real numbers. The requirements say nothing about real numbers, and a cellular automaton is the opposite extreme, the simplest dynamical system that can still meet them. We consider a chain of spins, each of which is up or down, written as 1 or 0. In discrete time steps every spin looks at

itself and its two neighbours and takes a new value given by a fixed table, the same table for every site. Such a system is a cellular automaton, a lattice model in which space, time and the state of a site are all discrete. Its update is a logical rule, and the automaton is the discrete counterpart of a spatially extended dynamical system such as a discretised wave or diffusion equation.

A cellular automaton is a candidate reservoir for two reasons. A table with only three input bits already produces every kind of behaviour met so far. Configurations freeze, patterns repeat, or one flipped bit spreads across the chain [27]. The spreading is the discrete counterpart of the growth of small differences that a spectral radius above one can cause in the echo state network. The table then decides, as the spectral radius did, whether the automaton forgets its past or amplifies it. The update is also a logical operation on bits, and logical operations such as AND multiply bits and supply the nonlinearity required of a reservoir. On digital hardware the update costs a table lookup or a few bit operations per cell. We test whether such a reservoir of bits, with the same linear readout, can forecast the records.

9.1 Cells, neighbours and a local rule

Cellular automaton

A cellular automaton consists of a ring of L cells $j = 0, 1, \ldots, L-1$, in which cell $L-1$ is the left neighbour of cell 0. Each cell has the state 0 or 1, and a local rule $f\colon \{0,1\}^3 \to \{0,1\}$ returns the new state of a cell from the states of the cell and its two neighbours. The state of cell j after i iterations is $c_j^{(i)}$, and the configuration is $\mathbf{c}^{(i)} = (c_0^{(i)}, \ldots, c_{L-1}^{(i)})$. The global map F applies the rule to all cells at once,

$$c_j^{(i)} = f\big(c_{j-1}^{(i-1)}, c_j^{(i-1)}, c_{j+1}^{(i-1)}\big) \quad \text{for all } j, \qquad \mathbf{c}^{(i)} = F\big(\mathbf{c}^{(i-1)}\big). \tag{9.1}$$

Every cell reads three cells of the previous configuration, all cells are updated together (Figure 22), and a signal travels at most one cell per iteration. Automata with two states and three cells in the neighbourhood are called elementary [27].

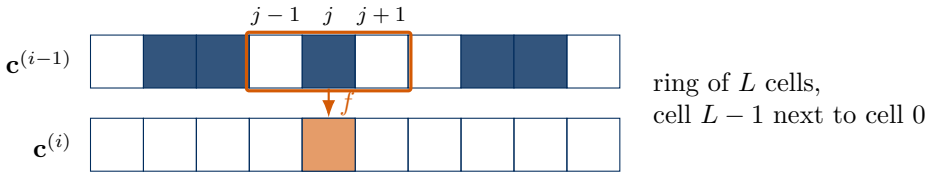


Figure 22: One update of an elementary cellular automaton (white 0, dark 1). The new state of cell j (orange) is the rule f applied to the three framed cells.

9.2 A single update and the rule number

A neighbourhood of three binary cells is one of the $2^3 = 8$ patterns $111, 110, 101, 100, 011, 010, 001, 000$, and a rule is a table that assigns a new state to each of them. Rule 30 has the table

pattern $c_{j-1}c_jc_{j+1}$	111	110	101	100	011	010	001	000
new state c_j	0	0	0	1	1	1	1	0

We take a ring of eight cells in the configuration $\mathbf{c}^{(0)} = 0\,1\,1\,0\,1\,0\,0\,0$. Cell 0 sees its left neighbour, cell 7 in state 0, itself in state 0 and cell 1 in state 1. For this pattern, 001, the table gives the new state 1. Cell 1 sees 011 and becomes 1, cell 2 sees 110 and becomes 0, and so on around the ring (Figure 23). The eight patterns are $001, 011, 110, 101, 010, 100, 000, 000$, and the new configuration is

$$\mathbf{c}^{(1)} = 1\,1\,0\,0\,1\,1\,0\,0.$$

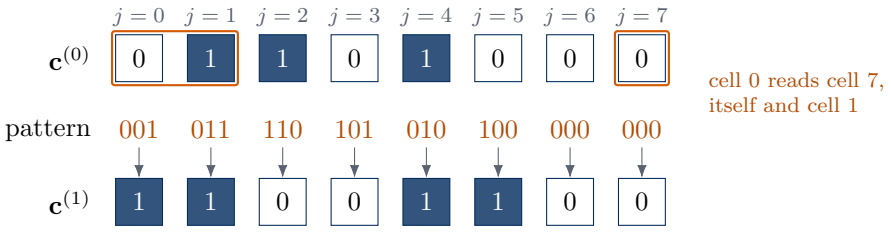


Figure 23: One update of rule 30 on a ring of eight cells. Top, $\mathbf{c}^{(0)}$. Middle, the pattern $c_{j-1}c_jc_{j+1}$ that each cell reads, with cell 0 reading cell 7 (orange frames). Bottom, the new states.

The bottom row of the table, read as a binary number, names the rule,

$$00011110_2 = 16 + 8 + 4 + 2 = 30.$$

Every row of eight bits is a rule, so there are $2^8 = 256$ elementary rules, numbered from 0 to 255. Rule 90 has the bottom row $01011010_2 = 64 + 16 + 8 + 2 = 90$. Its table gives 1 when the two neighbours differ and 0 when they agree, whatever the centre. Rule 90 therefore sets every cell to the exclusive or of its neighbours, $c_j^{(i)} = c_{j-1}^{(i-1)} \oplus c_{j+1}^{(i-1)}$, the sum modulo 2. The same ring of eight cells becomes 1 1 1 0 0 1 0 0 under rule 90. Rule 30 also has a short formula, $c_{j-1} \oplus (c_j \vee c_{j+1})$ with $\vee$ the logical or, and Figure 24 shows both tables.

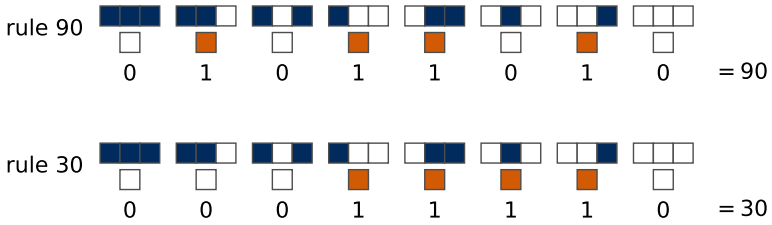


Figure 24: Rule tables of rules 90 and 30, the eight neighbourhood patterns (top rows) and the new state of the centre cell (bottom rows).

Iterating a rule produces a space-time diagram, one configuration per row. A single cell in state 1 on a ring of nine cells evolves under rule 90 as

$$\begin{array}{ll} i = 0 & 0\,0\,0\,0\,1\,0\,0\,0\,0 \\ i = 1 & 0\,0\,0\,1\,0\,1\,0\,0\,0 \\ i = 2 & 0\,0\,1\,0\,0\,0\,1\,0\,0 \\ i = 3 & 0\,1\,0\,1\,0\,1\,0\,1\,0 \\ i = 4 & 1\,0\,0\,0\,0\,0\,0\,0\,1 \end{array}$$

At $i = 2$ the centre is 0, since $1 \oplus 1 = 0$, and on a larger ring the pattern grows into nested triangles. Rule 90 is linear modulo 2. The new configuration is a sum of shifted copies of the old one, so superposition holds, $F(\mathbf{c} + \mathbf{c}') = F(\mathbf{c}) + F(\mathbf{c}')$ with the sum taken cell by cell modulo 2. Rule 150, $c_j \leftarrow c_{j-1} \oplus c_j \oplus c_{j+1}$, is linear in the same way.

9.3 Four classes of behaviour

The behaviour of automata started from random configurations falls into four classes [28] (Figure 25).

In class I every configuration dies out and the diagram turns white, as for rule 32. Its number is 00100000_2, so only the pattern 101 produces a 1, and a 1 is created only at a 0 between two 1s. An alternating stretch 1 0 1 0 ... keeps its pattern, shifted by one cell, and loses one cell at each end per

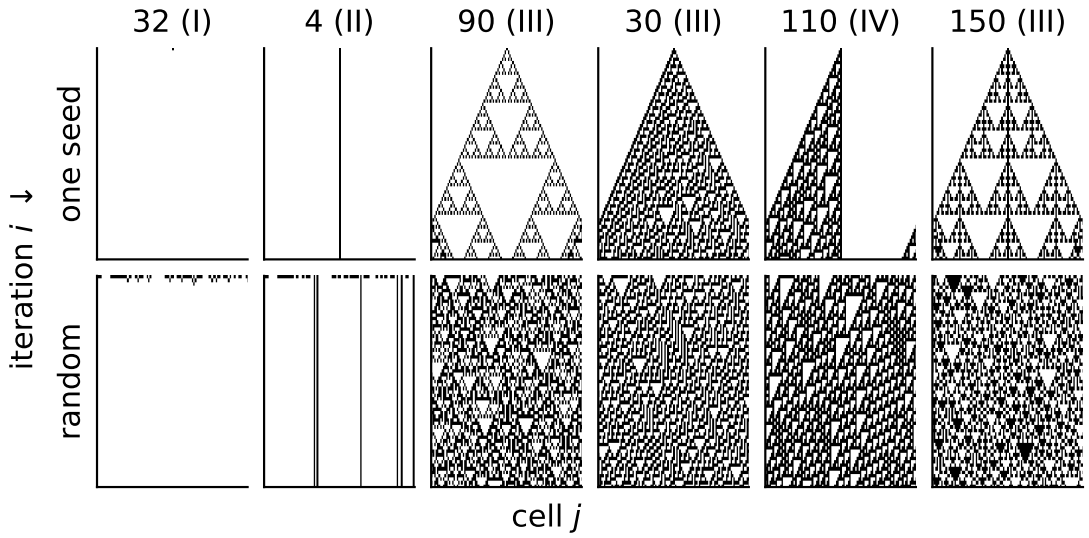


Figure 25: Space-time diagrams of six elementary rules on a ring of 101 cells, Wolfram class in brackets, iteration i downwards, black for state 1. Top, from a single cell in state 1. Bottom, from a random configuration.

iteration. On a ring of 14 cells,

$$\begin{array}{ll} i=0 & 01010101100100 \\ i=1 & 00101010000000 \\ i=2 & 00010100000000 \\ i=3 & 00001000000000 \\ i=4 & 00000000000000 \end{array}$$

From random configurations the lattice also empties within a few iterations.

In class II the configuration freezes or repeats, and the diagram shows vertical lines. Rule 4 keeps a 1 only in the pattern 010, so isolated cells in state 1 survive forever and everything else is erased after one iteration. In class III the diagram looks random. The linear rules 90 and 150 draw nested triangles from a single cell, rule 30 an irregular triangle. From a random start all three fill the lattice with patterns that do not repeat. In class IV localised structures move along the lattice on a regular background and collide, and rule 110 is the standard example, with its diagonal stripes in the figure.

9.4 Automata as physical models

Cellular automata were introduced by von Neumann as abstract machines that build copies of themselves [30]. In physics they serve as models in which a few local rules reproduce collective behaviour. Particles that hop and collide on a lattice obey, on large scales, the Navier–Stokes equation of a fluid [31], and cars on a lattice road reproduce the jams of freeway traffic [32]. The Game of Life [29] is the best-known automaton on a square lattice.

Rule 184 is the simplest model of traffic on a one-lane ring road. A cell in state 1 holds a car and a cell in state 0 is empty. The rule number $184 = 10111000_2$ lists the new state for the neighbourhoods $111, 110, \ldots, 000$ (Table 2). A cell becomes 1 if a car arrives from the left ($c_{j-1} = 1$, $c_j = 0$) or if its car is blocked and stays ($c_j = c_{j+1} = 1$). Each car therefore moves one cell to the right when the cell ahead is empty, and cars are never created or destroyed. The car density $\eta := n_{\mathrm{car}}/L$, the number n_{car} of cars divided by the number L of cells, is conserved by the rule. The flux Φ is the number of cars that move forward in one step, divided by L, the traffic current per cell. Rule 184 is the deterministic limit with maximum speed 1 of a stochastic traffic model [32].

Below half filling every car eventually finds an empty cell ahead and then moves at every step, so $\Phi = \eta$ (Figure 26, left). Above half filling the roles exchange. An empty cell with a car on its left is filled at the next step. The empty cells therefore move to the left like cars, at density $1 - \eta$ below one

111	110	101	100	011	010	001	000
1	0	1	1	1	0	0	0
blocked	moves on	arrives	arrives	blocked	moves on	stays empty	stays empty

Table 2: Rule 184 read as traffic. Top, the neighbourhoods (c_{j-1}, c_j, c_{j+1}) and the new state of cell j, the binary digits of 184. Bottom, what happens to cell j. A car is blocked and stays, a car moves on and leaves the cell empty, a car arrives from the left, or an empty cell stays empty.

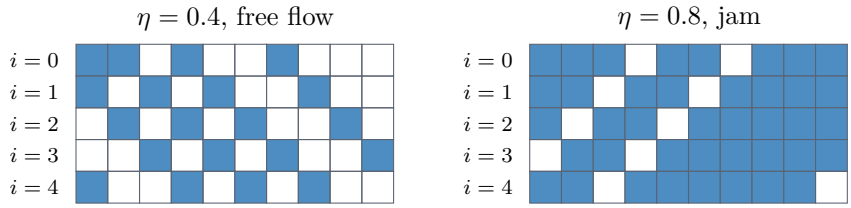


Figure 26: Rule 184 on a ring of ten cells, cars in blue moving to the right. Left, four cars, all moving after the first step. Right, eight cars, of which only the two directly behind the empty cells move.

half. Each of their moves is the move of one car, so $\Phi = 1 - \eta$ (Figure 26, right). Together,

$$\Phi(\eta) = \min(\eta,\, 1 - \eta), \tag{9.2}$$

with a cusp at $\eta = 1/2$, the transition between a free-flow phase and a jammed phase in which queues travel backwards. The flux measured on large rings follows this curve (Figure 27).

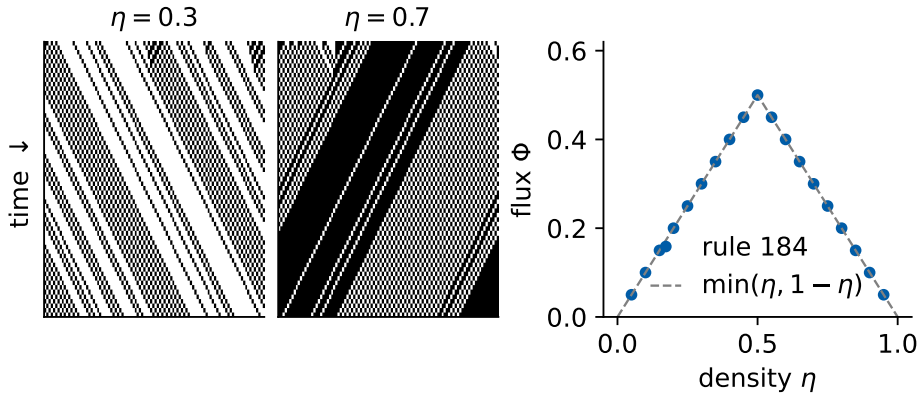


Figure 27: Rule 184. Left, space-time diagrams at $\eta = 0.3$ and $\eta = 0.7$. Right, the measured flux Φ against the density, with the curve of equation (9.2).

9.5 Damage spreading

Two copies, as in the test of the echo state property, also probe an automaton without input. We flip one cell of a configuration, iterate both copies and follow the set of cells in which they differ, the damage. We take the ring of nine cells 101100101 and flip its centre, cell 4. The rows below are the damage, a 1 wherever the two copies differ,

	rule 90	rule 30	rule 4
$i = 0$	000010000	000010000	000010000
$i = 1$	000101000	000011000	000000000
$i = 2$	001000100	000111100	000000000
$i = 3$	010101010	000010010	000000000

Under rule 90 the damage is the evolution of a single cell computed above, because superposition holds. The flipped copy is $\mathbf{c} + \mathbf{p}$, with $\mathbf{p}$ a configuration with a single 1, and $F^i(\mathbf{c} + \mathbf{p}) = F^i(\mathbf{c}) + F^i(\mathbf{p})$, so the difference $F^i(\mathbf{p})$ does not depend on the background. Under rule 30 the damage grows mostly

to the right. In the formula $c_{j-1} \oplus (c_j \vee c_{j+1})$ a difference in the left neighbour always changes the new state, while a difference in the right neighbour does so only when $c_j = 0$. Under rule 4 the damage disappears after one iteration, since in neither copy does a cell near the flip see the only surviving pattern 010. A flip that creates or removes an isolated 1 would instead stay on its cell forever.

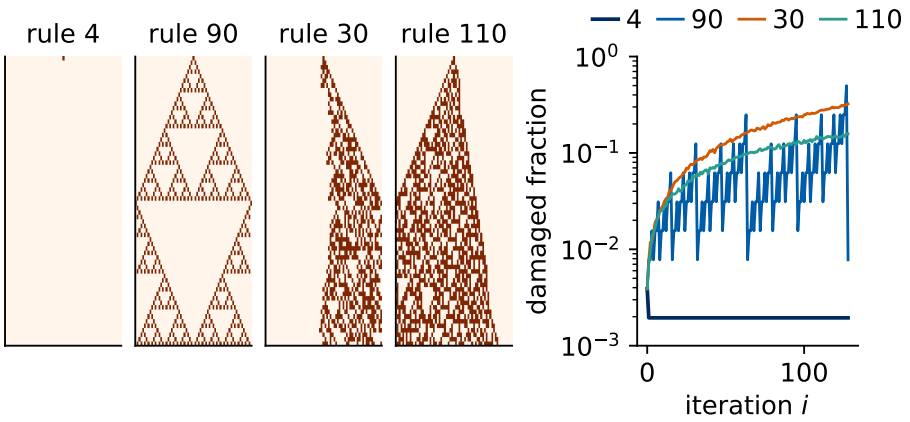


Figure 28: Damage spreading. Left, the damage (orange) for rules 4, 90, 30 and 110. Right, the damaged fraction of the ring against the iteration, averaged over random starts.

On a large ring and from random starts the three behaviours persist (Figure 28). Under rule 4 a difference stays on its cell or disappears, and under rule 110 it grows slowly and irregularly. Under rules 90 and 30 it opens into a cone that covers a large part of the ring. Since a cell is either 0 or 1, the difference cannot grow in size, and a cellular automaton shows its sensitivity to a perturbation in the extent of the damage. As an analogy, a growing cone plays the role of a positive Lyapunov exponent of a chaotic system. A damage that stays local or travels without growing plays the role of a stable or neutral direction. In a reservoir, a rule that spreads damage mixes every bit with its neighbours and scrambles a record of the input. A rule that transports a bit keeps it readable, and a rule that erases it keeps nothing.

A lone leaky unit with a common drive shrinks every difference by the factor $1 - \alpha$ per step, and an automaton has no such factor. A difference survives, travels or spreads unless the rule happens to erase it, as rule 4 did above. In a reservoir the input is written into some cells at every step, and a difference that reaches such a cell is overwritten in both copies. This overwriting is the only general mechanism by which a driven automaton can forget, and the echo state property of an automaton reservoir rests on it.

10 A cellular automaton as a reservoir

10.1 Echo state network and automaton compared

The random network of an echo state network can be replaced by a cellular automaton [33], and Figure 29 shows how the parts correspond. The readout of equation (2.1) and its ridge fit, equation (2.6), stay unchanged, with the bits read as the real numbers 0 and 1.

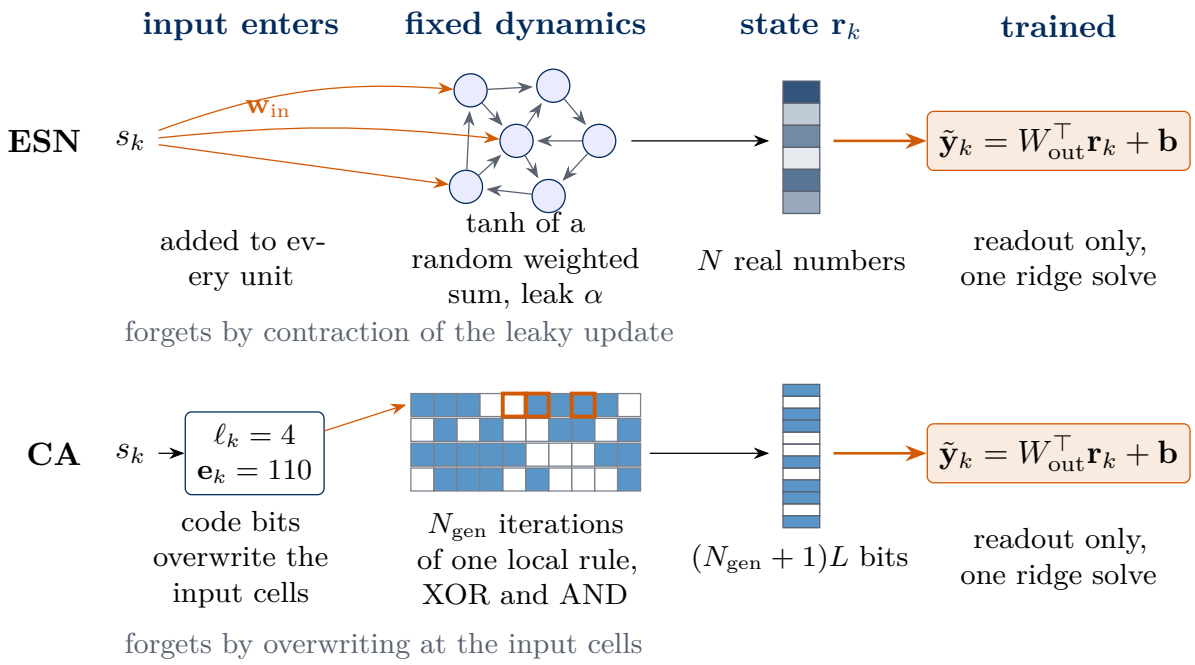


Figure 29: The echo state network (ESN, top) and the cellular automaton reservoir (CA, bottom), aligned part by part. The input enters as a weighted sum added to every unit or as code bits written into framed input cells. The fixed dynamics is a random network of hyperbolic tangents or N_{gen} iterations of one local rule. The state $\mathbf{r}_k$ is a column of N real numbers or of $(N_{\text{gen}} + 1)L$ bits, and in both only the linear readout is trained.

10.2 A driven ring of twelve cells

A ring of 12 cells shows how a driven automaton holds a record of its past inputs and forgets its start. The ring starts empty, and at every input step k a bit b_k is written into the first cell, replacing its content, after which the rule is applied once. Figure 30 shows the configurations for the input $1, 0, 1, 1, 0, 0, 1, 0, 0, 0$ under three rules.

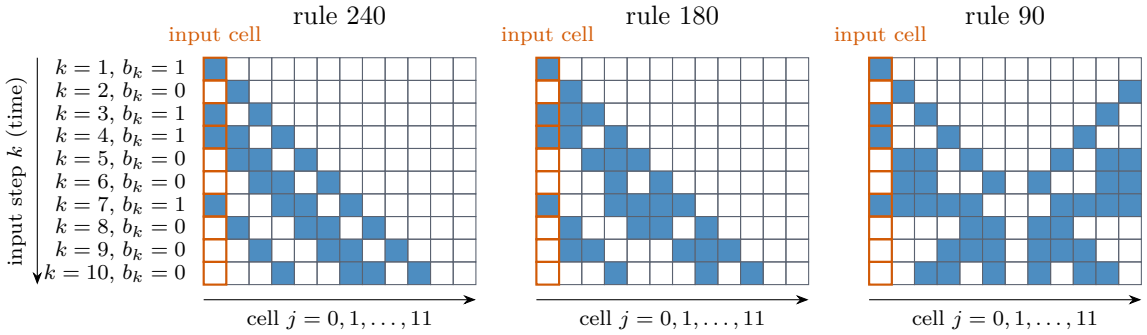


Figure 30: A driven ring of 12 cells, time running down and the cell index j across. Each row is the ring right after the bit b_k is written into cell 0, the input cell (orange frame). Row k therefore shows how the bits $b_k, b_{k-1}, \ldots$ sit along the ring, with cells in state 1 in blue. Left, rule 240, $c_j \leftarrow c_{j-1}$. Middle, rule 180. Right, rule 90.

Rule 240 copies the left neighbour, so at step k cell j holds the bit written j steps earlier. The row at $k = 7$, $1\,0\,0\,1\,1\,0\,1\,0\,\ldots$, is the input $b_7, b_6, \ldots, b_1$ read backwards. The configuration is a delay line like the shift reservoir of Figure 13, and a linear readout recovers b_{k-d} from cell d. After 12 steps a bit returns to the first cell and is overwritten by the new input. The ring holds the last 12 inputs and forgets everything older, a fading memory that ends abruptly at a fixed age. Overwriting also gives the echo state property, since two copies started from different configurations and driven by the same bits become identical after at most 12 steps.

When the cells hold the coded values of $s_k, s_{k-1}, \ldots$, a linear readout of them is an autoregressive

model on these coded values. A rule that also multiplies neighbouring cells adds products of bits of different ages, the kind of terms that NVAR adds to AR. Rule 180, $c_{j-1} \oplus c_j \oplus c_j c_{j+1}$, moves bits to the right and changes them on the way through the product $c_j c_{j+1}$. Rule 90 sends every bit in both directions, and copies of different inputs meet and add modulo 2, which a linear readout cannot undo.

For a memory test we drive the ring by a long sequence of random bits and take as features the configurations before and after the iteration, $\mathbf{c}_k^{(0)}$ and $\mathbf{c}_k^{(1)}$. For every delay d a least-squares readout returns b_{k-d}, scored by the recall capacity $C(d)$ of equation (6.1). Rule 240 recalls every past bit still on the ring without error and nothing older, and rule 180 recalls them partly. Rule 30 recalls little, and rules 90, 32 and 4 nothing but the present bit (Table 3). A readable record of the recent input needs a rule that moves bits along the lattice without spreading and mixing them.

rule	$C(1)$	$C(2)$	$C(3)$	$C(5)$	$C(8)$	$C(11)$	$C(12)$	MC
240	1.00	1.00	1.00	1.00	1.00	1.00	0.00	11.0
180	0.60	0.59	0.88	0.60	0.86	0.20	0.00	7.2
30	0.24	0.08	0.06	0.01	0.00	0.00	0.00	0.4
90, 32, 4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.0

Table 3: Recall capacity $C(d)$ of past bits for the driven ring of Figure 30. MC is the sum of $C(d)$ over $d \geq 1$. All rules recall the present bit, $C(0) = 1$.

10.3 One input step

The reservoir generalises the twelve-cell ring and runs on two clocks. The input step k advances by one sample s_k of the record. Inside one input step the rule is applied N_{gen} times, and the generations $i = 0, 1, \ldots, N_{\text{gen}}$ count these applications, with $i = 0$ the configuration right after the input is written. The ring of $L = RL_c$ cells is divided into R compartments, consecutive segments of L_c cells each. One input step consists of three operations, an encoding, an evolution and a readout, which Figure 31 shows on a small reservoir.

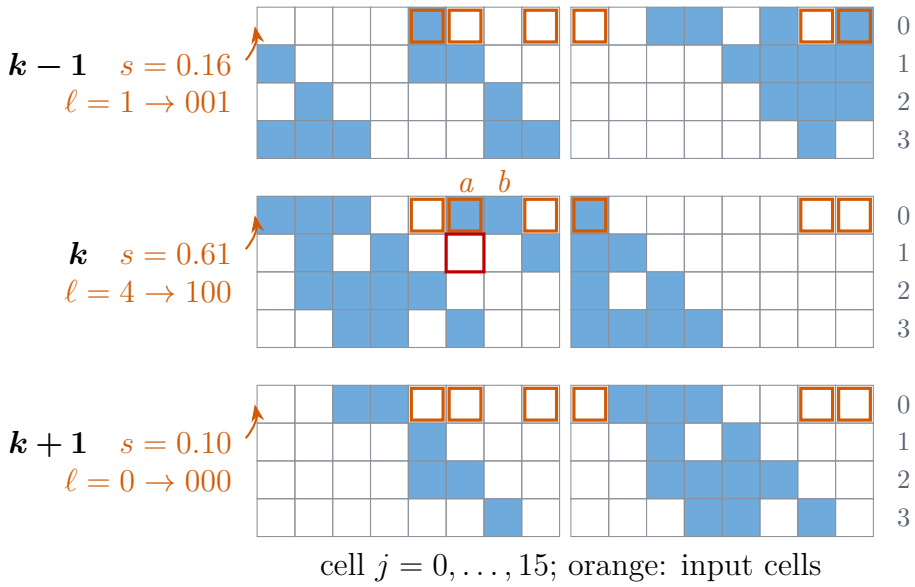


Figure 31: Three input steps of a reservoir with rule 180, $R = 2$ compartments of $L_c = 8$ cells, $Q = 8$ levels ($B = 3$ bits) and $N_{\text{gen}} = 3$. Left, the input step k (outer clock), the sample, its level and its binary code word, written into the input cells (orange) of row $i = 0$. Right of each block, the generation i (inner clock). The $N_{\text{gen}} + 1$ rows of step k form the feature vector $\mathbf{r}_k$ of the readout. Labels a, b and the red frame mark the traced cell of the text.

10.4 Encoding

An automaton holds bits, so each sample $s_k \in [0,1]$ is first rounded [34]. The interval $[0,1]$ is cut into Q bins of width $1/Q$, the quantisation levels, and the sample becomes the number of its bin,

$$\ell_k = \mathrm{clip}(\lfloor Q s_k \rfloor, 0, Q-1), \tag{10.1}$$

where clipping handles test values outside $[0,1]$. For $Q = 8$ the sample $s_k = 0.61$ lies in the bin from 0.5 to 0.625, so $\ell_k = 4$ (Figure 32).

The level is written in binary as a code word $\mathbf{e}_k$ of $B = \log_2 Q$ bits, so $Q = 8$ levels need $B = 3$ bits, and the forecasting reservoirs use $Q = 16$ and $B = 4$. The sample $s_k = 0.61$ with level $\ell_k = 4$ thus becomes the code word $\mathbf{e}_k = 100$.

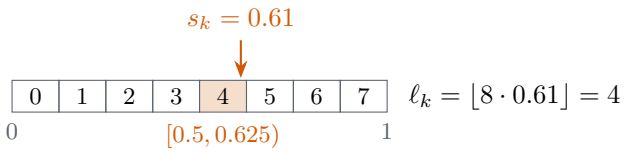


Figure 32: Quantisation of one sample. The interval $[0,1]$ cut into $Q = 8$ bins of width $1/8$ and the sample $s_k = 0.61$ in bin 4.

In each compartment B input cells are chosen once at random and kept fixed for all steps, so the R compartments hold R independent placements of the same code word. The bits of the code word overwrite these cells in the configuration left by the previous step. All other cells keep their content, what the earlier inputs and the rule left there, and this kept content is the memory of the reservoir. In Figure 31 the sample of step k is again $s_k = 0.61$ with the code word 100. The row $i = 0$ of step k differs from the row above it only in input cells. A larger Q rounds less but overwrites more cells.

10.5 Evolution

Between two inputs the rule is applied N_{gen} times to the whole ring, $\mathbf{c}_k^{(i)} = F(\mathbf{c}_k^{(i-1)})$, across the compartment boundaries as well. Rule 180 updates every cell as

$$c_j \leftarrow c_{j-1} \oplus c_j \oplus c_j c_{j+1}, \tag{10.2}$$

with two bit operations, the exclusive or $\oplus$ (XOR) and the logical AND, written as the product $c_j c_{j+1}$. Table 4 lists them on all pairs of bits. XOR is addition modulo 2, 1 when exactly one input is 1, and AND is the ordinary product of two bits, 1 only when both are 1.

a	b	$a \oplus b$ (XOR)	ab (AND)	$a \oplus ab$
0	0	0	0	0
0	1	1	0	0
1	0	1	0	1
1	1	0	1	0

c_{j-1}	c_j	c_{j+1}	new c_j by equation (10.2)
0	1	1	$0 \oplus 1 \oplus 1 \cdot 1 = 0$
0	1	0	$0 \oplus 1 \oplus 1 \cdot 0 = 1$

Table 4: Left, the bit operations of the rules. The last column is 1 only for $a = 1$, $b = 0$, the logical a and not b, which in ordinary arithmetic is $a(1-b) = a - ab$. Right, two cell updates of rule 180. They agree with the rule table $180 = 10110100_2$, whose digits are the new states for the neighbourhoods $111, 110, \ldots, 000$, here the entries 0 for 011 and 1 for 010.

The rule moves the new bits away from the input cells, on average one cell to the right per generation. On the way it combines them with the bits of earlier inputs that they meet. Figure 33 traces one such meeting in step k of Figure 31. In the row $i = 0$ the input cell $j = 5$ has just received the most significant bit of s_k, $a = 1$. Its right neighbour holds $b = 1$, a bit that depends on s_{k-1}, and

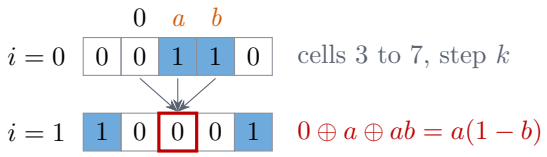


Figure 33: Cells 3 to 7 of step k at generations 0 and 1. The input cell 5 holds the bit a of s_k, its right neighbour the bit b left by s_{k-1}, its left neighbour 0. One generation of rule 180 writes $0 \oplus a \oplus ab = a(1-b)$ into it (red frame).

its left neighbour is 0. The neighbourhood $(0, 1, 1)$ is the first update of Table 4, so one generation later the cell holds $0 \oplus a \oplus ab = a(1-b) = 0$, a product of a bit of s_k and a bit of s_{k-1}. Its neighbours in this row update likewise,

cell j	3	4	5	6	7
(c_{j-1}, c_j, c_{j+1}) at $i = 0$	$(1,0,0)$	$(0,0,1)$	$(0,1,1)$	$(1,1,0)$	$(1,0,1)$
$c_{j-1} \oplus c_j \oplus c_j c_{j+1}$	$1 \oplus 0 \oplus 0$	$0 \oplus 0 \oplus 0$	$0 \oplus 1 \oplus 1$	$1 \oplus 1 \oplus 0$	$1 \oplus 0 \oplus 0$
c_j at $i = 1$	1	0	0	0	1

After N_{gen} generations the ring is a record of the recent input history that also contains products of past bits.

10.6 Readout and training

The $N_{\text{gen}} + 1$ configurations of one input step, the row $i = 0$ after the writing and the rows after each generation, are concatenated into the feature vector

$$\mathbf{r}_k = (\mathbf{c}_k^{(0)}, \mathbf{c}_k^{(1)}, \ldots, \mathbf{c}_k^{(N_{\text{gen}})}) \in \{0,1\}^{(N_{\text{gen}}+1)L}, \tag{10.3}$$

64 bits in Figure 31, read as the real numbers 0 and 1. The forecast of the next 50 values is the readout of equation (2.1), $\tilde{\mathbf{y}}_k = W_{\text{out}}^{\top} \mathbf{r}_k + \mathbf{b}$. The weights W_{out} and the offset $\mathbf{b}$ are fitted once by the ridge regression of equation (2.6) over all training steps, and they are the only trained numbers. The ridge strength and the settings of the automaton are chosen on the validation block. The last configuration $\mathbf{c}_k^{(N_{\text{gen}})}$ is carried over and receives the code word of s_{k+1}, so step $k+1$ starts where step k ends.

10.7 One input step with numbers

We take a smaller reservoir with $Q = 8$ levels, code words of $B = 3$ bits and $R = 2$ compartments of $L_c = 5$ cells, so a ring of $L = 10$ cells. It runs $N_{\text{gen}} = 2$ generations of rule 90, $c_j \leftarrow c_{j-1} \oplus c_{j+1}$ on the ring. The input cells are $\{0, 2, 3\}$ in the first compartment and $\{6, 7, 9\}$ in the second, drawn once and kept for all steps. Figure 34 draws the whole chain.

The present value is $s_k = 0.61$. Equation (10.1) gives the level $\ell_k = \lfloor 8 \cdot 0.61 \rfloor = \lfloor 4.88 \rfloor = 4$. The level 4 is the code word 100 in binary. The previous step left the configuration

$$\mathbf{c}_{k-1}^{(2)} = 1\,0\,0\,1\,1 \,|\, 0\,1\,0\,0\,1,$$

with a bar between the compartments. The three code bits $1, 0, 0$ replace the contents of cells $0, 2, 3$ and of cells $6, 7, 9$ in this order, so that cells 0 and 6 receive the most significant bit. All other cells keep their states,

$$\mathbf{c}_k^{(0)} = 1\,0\,0\,0\,1 \,|\, 0\,1\,0\,0\,0.$$

Two generations of rule 90 follow. Cell 0 becomes $c_9 \oplus c_1 = 0 \oplus 0 = 0$, cell 1 becomes $c_0 \oplus c_2 = 1 \oplus 0 = 1$, cell 5 becomes $c_4 \oplus c_6 = 1 \oplus 1 = 0$, and so on around the ring,

$$\mathbf{c}_k^{(1)} = 0\,1\,0\,1\,0 \,|\, 0\,0\,1\,0\,1, \qquad \mathbf{c}_k^{(2)} = 0\,0\,0\,0\,1 \,|\, 0\,1\,0\,0\,0.$$

The feature vector is the three rows side by side, $\mathbf{r}_k = (\mathbf{c}_k^{(0)}, \mathbf{c}_k^{(1)}, \mathbf{c}_k^{(2)})$, thirty numbers 0 or 1. The last row $\mathbf{c}_k^{(2)}$ receives the code word of s_{k+1} at the next step. Its cells 1, 4, 5 and 8 carry into step $k+1$ what s_k and the earlier values left behind.

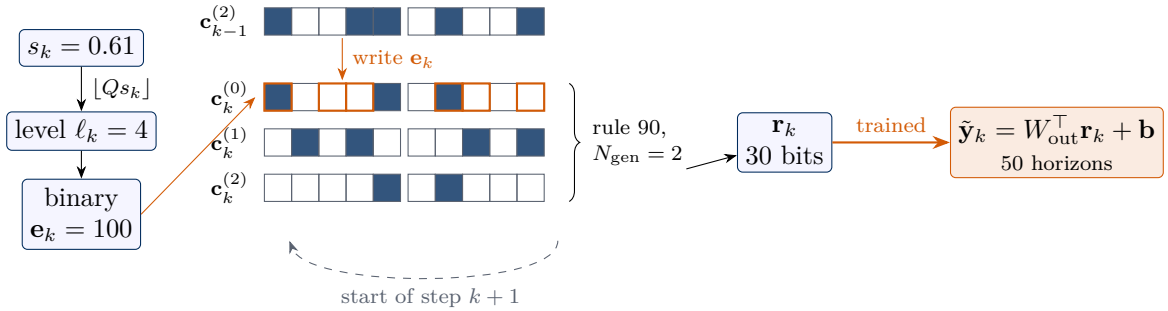


Figure 34: The input step of the text. Input cells framed in orange, two compartments of five cells separated by a small gap. Only the linear readout (orange arrow) is trained.

10.8 The number of generations

The number of generations per input step is a hyperparameter, chosen on the validation block together with the ridge strength β from $N_{\text{gen}} \in \{1, 2, 3, 4, 6, 8\}$. Under a transport rule such as 180 a bit moves about one cell per generation. Between two writings the record therefore shifts by about N_{gen} cells, and consecutive inputs lie about N_{gen} cells apart. The trail behind the input cells of one compartment then holds roughly L_c/N_{gen} past inputs before it runs into the next compartment, where it meets newer bits. A bit that sits on an input cell at a writing is erased. N_{gen} thus trades the spacing of the record against its length and also sets the number of features, $(N_{\text{gen}}+1)RL_c$. Under a spreading rule such as 90 or 30 every generation mixes each bit with more of its neighbours, and many generations scramble the record. The pure shift, rule 170, $c_j \leftarrow c_{j+1}$, which moves every bit one cell to the left per generation and never combines bits, serves as a reference.

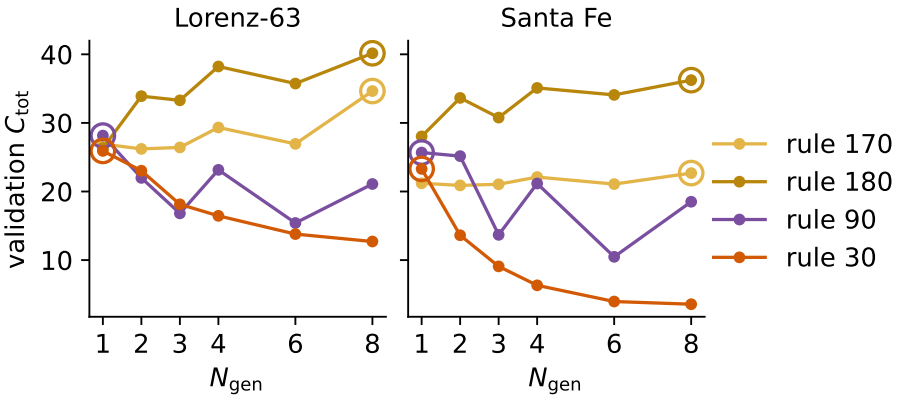


Figure 35: Validation C_{tot} against N_{gen} for the pure shift 170, rule 180 and the spreading rules 90 and 30, on Lorenz (left) and the laser (right). The reservoir has $Q = 16$, $R = 16$ compartments of $L_c = 16$ cells and one placement of the input cells. Open circles, the selected N_{gen}.

Figure 35 shows both behaviours. Rule 180 gains with more generations, though not at every step of the grid, and selects one of the larger values on both records. The pure shift hardly depends on N_{gen}, since each of its generations is a shifted copy of the configuration written at $i = 0$ and adds the same bits again. The spreading rules lose with more generations.

10.9 Nonlinearity from logic on bits

The readout only adds the bits with weights, so whatever is nonlinear in the forecast must already be in the bits. The traced cell of Figure 33 shows how it gets there. One generation of rule 180 writes $0 \oplus a \oplus ab = a(1-b) = a - ab$ into the cell. This function of the bit a of s_k and the bit b of s_{k-1} is 1 only for $a = 1$, $b = 0$ (Table 4). No weighted sum $w_0 + w_1 a + w_2 b$ reproduces it. The row $(0, 0)$ requires $w_0 = 0$, the rows $(1, 0)$ and $(0, 1)$ then require $w_1 = 1$ and $w_2 = 0$, and the row $(1, 1)$ gives $w_1 + w_2 = 1$ where the rule requires 0.

The readout multiplies the cell by a weight w, and the forecast contains the term $w\, a(1-b)$. Since each code bit is a step function of its sample, this term is a nonlinear function of the two inputs s_k and s_{k-1}. Such a term is the analogue of the product $s_k s_{k-1}$ that NVAR adds to an autoregressive model. An echo state network makes its nonlinearity from continuous sums passed through the hyperbolic tangent, and the automaton makes it with logic on bits. Even the exclusive or alone produces products for a linear readout, since for bits read as numbers $a \oplus b = a + b - 2ab$, as the rows of Table 4 confirm. Rule 90, without any product in its formula, therefore still multiplies bits, and rule 30, $c_{j-1} \oplus c_j \oplus c_{j+1} \oplus c_j c_{j+1}$, contains both.

The encoding supplies the step functions. In binary the level is the weighted sum of its code bits, $\ell_k = \sum_{b=1}^{B} 2^{B-b} e_{k,b}$ with $e_{k,1}$ the most significant bit. A linear readout of the bits of one sample therefore recovers the quantised sample. Functions of two or more samples, such as $s_k s_{k-1}$, need products of bits, which the rule supplies.

The pure shift 170, $c_j \leftarrow c_{j+1}$, only moves bits and never combines them, so its features are the staircases of past values side by side. Rule 180 moves the bits in the same way and adds one local product. Driven by independent inputs uniform in $[0, 1]$, both reservoirs recall the input three steps back about equally well (Figure 36). The products of two past inputs separate them. A sum of functions of single samples has zero correlation with the centred product $\delta s_{k-1}\, \delta s_{k-4}$, $\delta s := s - 1/2$, of independent inputs, and rule 170 recovers none of it. Rule 180 recovers a clear part of the product of consecutive inputs and a smaller part of the product of inputs three steps apart. The readout is the same linear ridge regression in both cases, and the gain comes from the products that the fixed rule forms between bits of different past inputs.

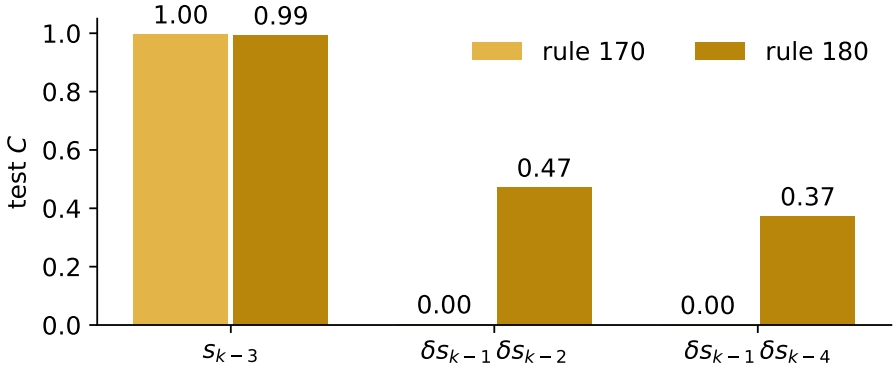


Figure 36: Test capacity C, the squared correlation of equation (6.1) between the readout and its target, of a ridge readout of the pure shift 170 and of rule 180 ($Q = 16$, $R = L_c = 16$, $N_{\text{gen}} = 8$). Both are driven by independent inputs s_k uniform in $[0, 1]$. The three targets are the delayed input s_{k-3} and the products $\delta s_{k-1}\, \delta s_{k-2}$ and $\delta s_{k-1}\, \delta s_{k-4}$, with $\delta s := s - 1/2$.

10.10 The echo state property with input

The input overwrites only B cells of each compartment at every step, so the other cells need not forget the initial configuration. The property therefore has to be tested with the input. Two copies of the forecasting reservoir start from different random configurations of the whole ring and receive the same encoded laser record. Under rule 180 and under rule 210, $c_j \leftarrow c_{j-1} \oplus c_{j+1} \oplus c_j c_{j+1}$, which also moves bits and multiplies neighbours, the copies become identical in every pair of starts within a few

tens of steps. Under rules 90, 30 and 4 they never become identical within the record. On the test origins a large fraction of the cells still differs under rules 90 and 30, and a smaller one under rule 4. Under the moving rules a difference travels until it reaches an input cell, where it is overwritten. Under rule 4 a difference on an isolated cell away from the input cells stays in place forever.

10.11 Three fates of an injected bit

Figure 37 follows a single bit on a quiet ring of 81 cells under nine rules. Above each panel the rule is written in its algebraic normal form, a sum modulo 2 of products of the three neighbourhood cells. Rule 180, $c_{j-1} \oplus (c_j \wedge \neg c_{j+1})$ with $\wedge$ the logical and and $\neg$ the negation, reads $c_{j-1} \oplus c_j \oplus c_j c_{j+1}$ in this form, since $\neg c_{j+1} = 1 \oplus c_{j+1}$.

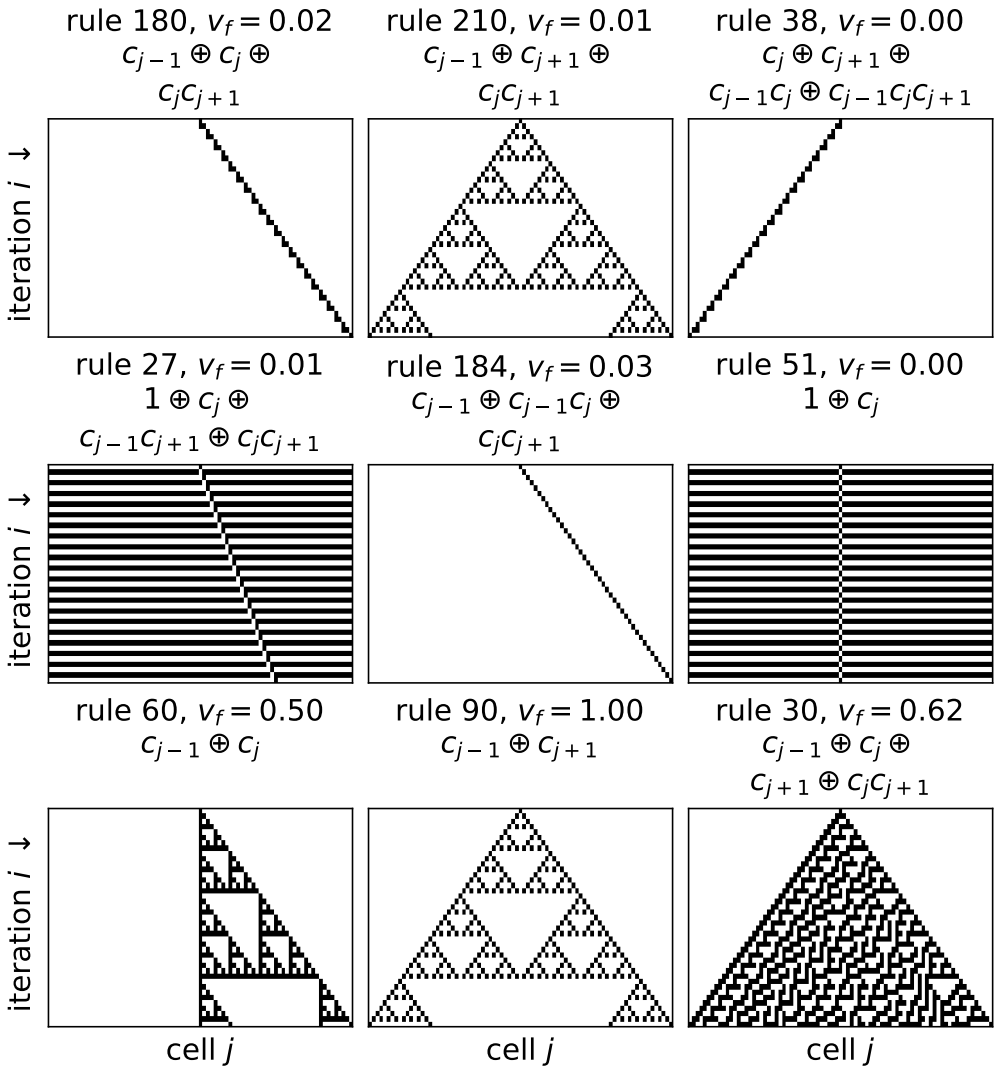


Figure 37: A single bit on a quiet ring of 81 cells over 40 iterations, iteration i downwards, black for state 1. Above each panel stand the rule number, the algebraic normal form and the label v_f. This label is the width of the damage left by one flipped cell on a random background, divided by its largest possible width.

Under rule 180 the bit travels to the right by one cell per iteration on average, alternating between a single bit and a pair of neighbouring bits. Rules 38 and 138 carry it to the left, and rule 27, $c_j \leftarrow 1 \oplus c_j \oplus c_j c_{j+1} \oplus c_{j-1} c_{j+1}$, moves it on a background that blinks between all 0 and all 1. A bit that travels remains a readable record of an earlier input. Under rules 90 and 60 the bit spreads into nested triangles and under rule 30 into an irregular triangle, and it mixes with the bits of later inputs. Under rule 204, $c_j \leftarrow c_j$, the bit stays where it was written until the next input overwrites it. A single bit on a quiet ring shows only part of the behaviour. Rule 210 draws the same triangles as rule 90 from one cell, because the 1s of that pattern are never neighbours and its product term $c_j c_{j+1}$

vanishes. On a random background its damage hardly spreads and drifts to the right by one cell per iteration.

10.12 Forecast capacity of representative rules

We run nine rules that span these behaviours with the forecasting protocol and one random placement of the input cells for every rule. The ridge strength and the number of generations are chosen on the validation block for every rule and record. The value of N_{gen}, and with it the number of features, therefore differs between rules and records. The nine rules are the identity 204, $c_j \leftarrow c_j$, the pure shift 170, $c_j \leftarrow c_{j+1}$, and three rules that move bits and add local products. These three are 180, 27 and the traffic rule 184 of Figure 26, $c_j \leftarrow c_{j-1} \oplus c_{j-1}c_j \oplus c_j c_{j+1}$, whose cars now carry input bits. The remaining four are the additive rules 90 and 150, the chaotic rule 30 and the complex rule 110. Figure 38 gives the test capacity $C(\tau)$ against the horizon, and Table 5 the total capacity on all five records.

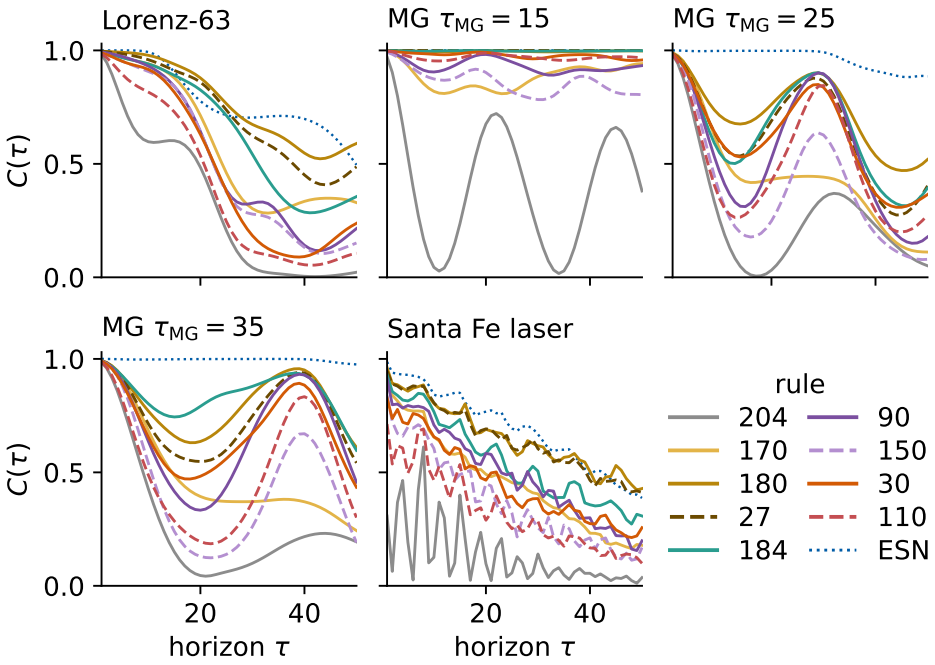


Figure 38: Test capacity $C(\tau)$ against the horizon τ for nine elementary rules, each at its selected N_{gen} ($Q = 16$, $R = L_c = 16$). Grey rule 204, gold and brown 170, 180 and 27, teal 184, purple 90 and 150, orange 30, red 110. Dotted blue, the echo state network with 300 units.

The identity rule 204 lets its readout see only the present quantised value, and it gives the lowest C_{tot} on every record, close to persistence. The pure shift 170, a delay line of coded past values like the twelve-cell ring under rule 240, lifts the Mackey–Glass curves far above the identity. On Lorenz it forecasts at the level of the autoregressive model, and rule 180, which adds local products of bits (Figure 36), reaches nearly three times that value. On the laser rule 180 also gains strongly over the pure shift.

Rules 180, 27 and 184 give the highest curves on Lorenz and the laser. Rule 180 is the best of the nine rules on four records. It comes close to the echo state network, slightly above it on Lorenz and slightly below it on the laser. On MG 25 and MG 35 it stays below the network.

The rules that spread damage, 90, 150, 30 and 110, stay below them on Lorenz and the laser. On almost every record these four rules select one or two generations. The order has exceptions, and on MG 25 rule 30 comes out above rules 27 and 170.

The curves follow the damage and echo state tests. Rule 204 has no memory to read, and rules 90 and 30 scramble the inputs and fail the echo state test. The moving rules carry the code words along the lattice, and rules 180 and 210 forget their start. Slow spreading alone is not sufficient, since rules

test C_{tot} (N_{gen})	Lorenz	MG 15	MG 25	MG 35	Santa Fe
rule 204	11.9 (1)	20.0 (1)	14.9 (8)	17.5 (8)	7.0 (1)
rule 170	14.3 (2)	47.1 (8)	34.1 (8)	36.0 (8)	21.3 (4)
rule 180	40.5 (8)	49.9 (6)	42.8 (8)	44.3 (8)	33.3 (4)
rule 27	35.8 (6)	50.0 (3)	34.7 (6)	41.8 (8)	32.8 (6)
rule 184	36.7 (8)	49.8 (8)	37.6 (1)	41.6 (1)	30.7 (2)
rule 90	24.6 (1)	47.1 (1)	24.7 (1)	30.0 (1)	24.2 (1)
rule 150	13.6 (1)	38.6 (2)	18.5 (2)	21.1 (2)	17.8 (2)
rule 30	22.5 (1)	45.0 (3)	35.5 (1)	38.3 (1)	22.7 (1)
rule 110	19.0 (2)	48.5 (2)	26.2 (1)	28.9 (1)	19.4 (1)
AR	15.4	49.8	33.1	38.0	16.5
NVAR	15.4	50.0	45.5	48.6	34.7
ESN, 300 units	39.2	50.0	48.3	49.8	34.7

Table 5: Test C_{tot} of nine elementary rules, with the selected N_{gen} in brackets, and the AR, NVAR and echo state network references.

204 and 170 spread no damage at all and forecast far worse than rule 180. The useful rules carry the input along the lattice and also form local products of it. These conclusions rest on one placement of the input cells, one quantisation and one lattice. The spread over placements leaves differences of one or two units between rules unresolved.

10.13 A map of all elementary rules

Reflecting a rule from left to right gives the rule $f'(c_{j-1}, c_j, c_{j+1}) = f(c_{j+1}, c_j, c_{j-1})$, whose dynamics is the mirror image of the original one. Exchanging the roles of 0 and 1 gives $f'(c_{j-1}, c_j, c_{j+1}) = 1-f(1-c_{j-1}, 1-c_j, 1-c_{j+1})$, whose dynamics is the original one with every configuration complemented. The 256 rules fall into 88 such classes of copies, each labelled by its smallest rule number. Rules 154, 166, 180 and 210 form one class, rule 210 being the complement of rule 180 and rule 166 its mirror image. Inside a reservoir the members of a class can still differ. The written code word is not complemented with the rule and a mirrored lattice meets a different random placement.

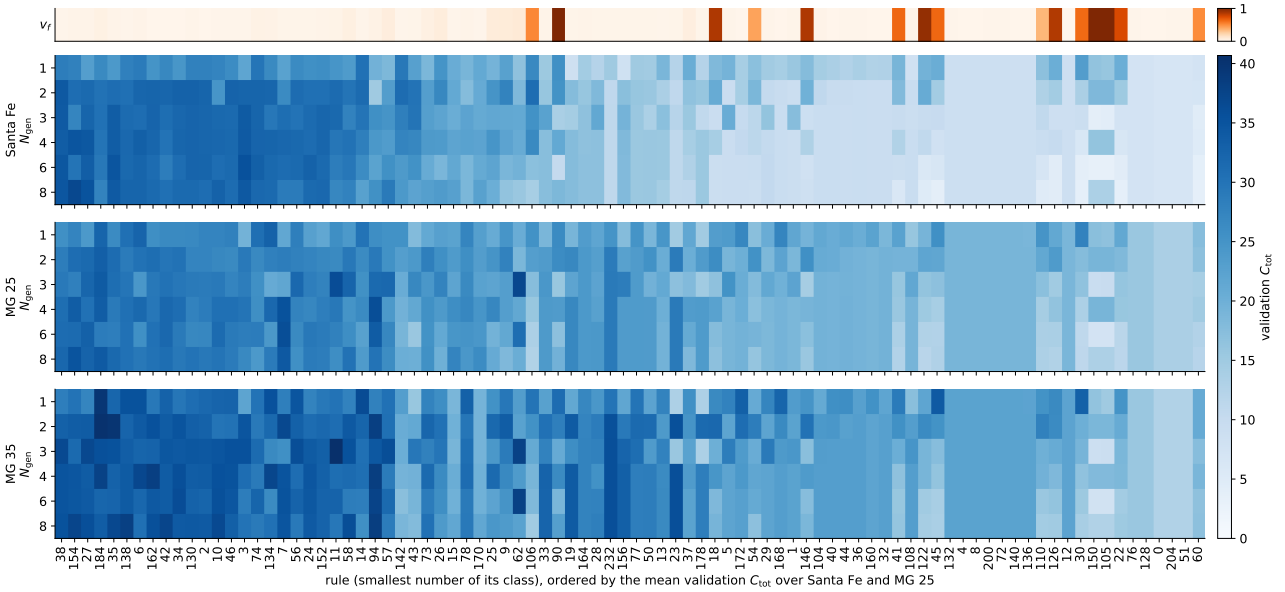


Figure 39: Validation C_{tot} for the 88 classes of elementary rules (columns, labelled by the smallest rule number) and N_{gen} (rows), for the laser (top), MG 25 and MG 35. Columns ordered by the mean over the laser, MG 25 and all N_{gen}, best on the left. Top strip, how far a single flipped cell spreads on a random background, from not at all (pale) to a full cone (dark).

Each entry of Figure 39 is the validation C_{tot} of one reservoir with the settings of the nine-rule comparison. The left third of the map is dark on all three records while the strip above it is pale, and all ten best classes spread damage slowly. The class of rule 27 is best by the mean. The class of rule 154 gives the best single entries on the laser and on MG 25, both at the largest N_{gen}. Near the right end lie rules 0, 51 and 204, whose values do not change with N_{gen}, since nothing of an earlier input survives the next writing. The three records rank the rules alike, with a Spearman rank correlation, the Pearson correlation of the ranks, of +0.91 between the laser and each Mackey–Glass record.

The rows show what N_{gen} does. For the moving rules of the class of rule 154 the laser score rises steadily from one to eight generations. For the spreading rules it falls, strongly for rule 30 and on average for the fast-spreading classes, while the average over the slowly spreading classes hardly changes.

Damage spreading and forecast quality are linked in one direction only. None of the fast-spreading rules is among the best classes. Every good reservoir in the map spreads damage slowly, while many slowly spreading rules, those that erase or freeze the input, make bad ones. Over five random placements of the input cells the order of the five best classes holds on the laser and is unresolved on MG 25. The number of features grows with N_{gen}, so part of the gain of the moving rules at large N_{gen} may come from the larger readout.

10.14 Forecasts of the selected automaton

The forecasting search compares eight elementary rules, 32, 4, 90, 150, 30, 110, 180 and 210, on the validation block. The search selects rule 180 for four records and rule 210 for MG 35, both members of the class of rule 154. A second search then sets the number of levels, the number of generations and the number of compartments. Adding the code word modulo 2 in place of replacing the cell contents gives far lower validation scores on the four irregular records. At every selected setting the two copies of the echo state test become identical, within a few tens of steps for $Q = 16$. For MG 15 with $Q = 4$ this takes longer than the washout, so there the first rows of the fit still depend on the start.

test C_{tot}	Lorenz	MG 15	MG 25	MG 35	Santa Fe
AR	15.4	49.8	33.1	38.0	16.5
NVAR	15.4	50.0	45.5	48.6	34.7
ESN, 300 features	39.2	50.0	48.3	49.8	34.7
automaton, 320 features	29.0	49.9	34.5	40.9	28.6
automaton, selected	40.5	50.0	42.8	44.8	33.3
rule	180	180	180	210	180
(Q, N_{gen}, R)	(16, 8, 16)	(4, 2, 16)	(16, 8, 16)	(16, 8, 16)	(16, 4, 16)
features	2304	768	2304	2304	1280

Table 6: Total capacity on the test block of AR, NVAR, the echo state network and the cellular automaton reservoir. The automaton is shown with 320 features ($R = 4$, $N_{\mathrm{gen}} = 4$) and at the setting selected on the validation block. The selection runs over $Q \in \{4, 16, 64\}$, $N_{\mathrm{gen}} \in \{1, 2, 4, 8\}$, $R \in \{4, 8, 16\}$ with $L_c = 16$.

On Lorenz the automaton slightly exceeds the echo state network, and on the laser it comes close, within the spread over random reservoirs. On MG 25 and MG 35 it stays about five units below (Table 6, Figure 40). The selected automata use up to nearly eight times as many features as the echo state network, and with 320 features they fall behind it on every irregular record.

The one-step forecasts show where the automaton loses (Figure 41). On the laser its one-step capacity stays visibly below that of the echo state network. The automaton never sees changes of the signal smaller than one level, 1/16 of the range at the selected Q. On these records the automaton does best where the forecast needs a long memory and coarse amplitude information, as for Lorenz, and falls behind where fine amplitude resolution matters.

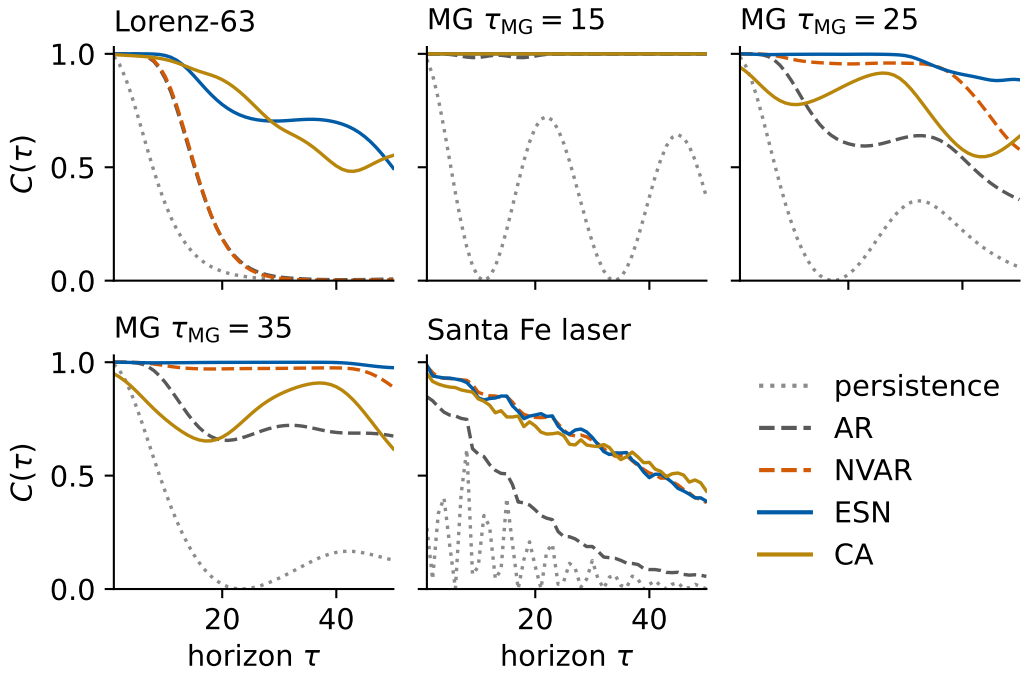


Figure 40: Test capacity $C(\tau)$ against the horizon for persistence, AR, NVAR, the echo state network and the selected cellular automaton reservoir (CA).

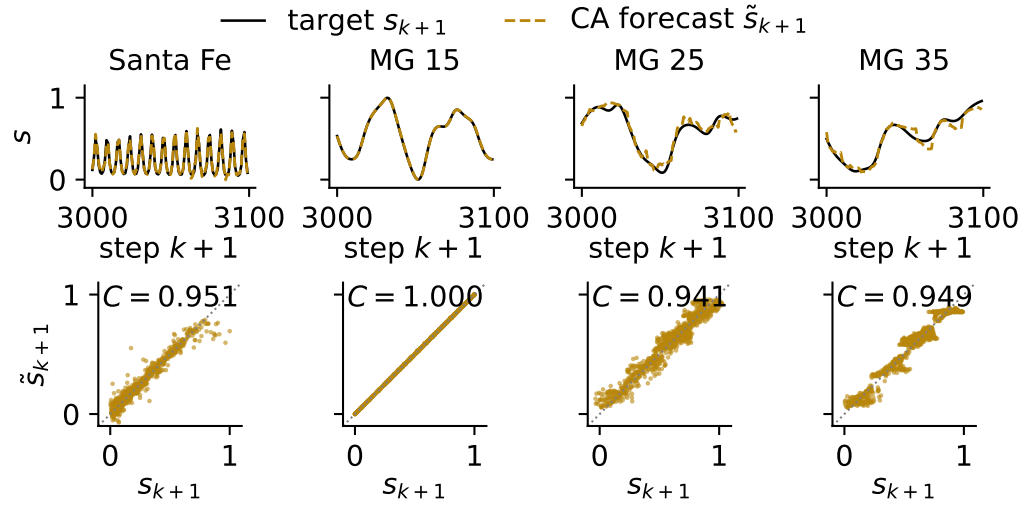


Figure 41: One-step forecasts of the cellular automaton reservoir. Top, the target s_{k+1} and the forecast for the first 100 test targets. Bottom, the forecast against the target, with the capacity C.

10.15 Transport and local products

The forecasts of the best rules may come from the transport of the written code bits alone, a delay line of coded past values, or from the local products. A control model without any automaton separates the two. Its feature vector holds the binary code bits, $Q = 16$ and $B = 4$, of the last n_{lag} samples, a shift register of code words, with the ridge readout of the automaton. The variants are a short register of $n_{\text{lag}} = 64$ lags and a long register with as many features as the selected automaton. A third register adds all 16 products of a bit of lag d with a bit of lag $d + 1$, with n_{lag} chosen to match the size of the automaton.

On MG 25 and MG 35 the short plain register beats the automaton, so there the transport of quantised past values accounts for the forecast (Figure 42). On Lorenz and the laser the plain register falls far behind. The products of adjacent lags bring it close to the automaton on Lorenz, while on the laser the automaton stays clearly ahead. On these two records products of neighbouring code bits are needed. The long register with as many features as the automaton does worse than the short one

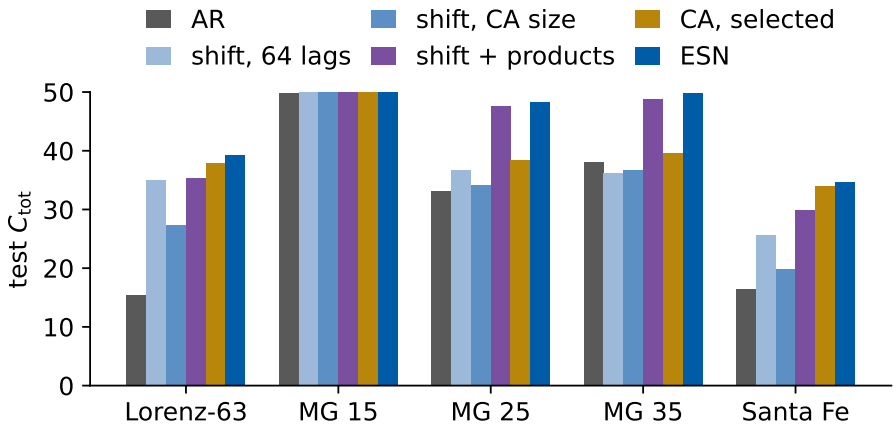


Figure 42: Test C_{tot} of AR, of the three shift registers, of the selected cellular automaton reservoir and of the echo state network.

on Lorenz, MG 25 and the laser.

11 A quantum reservoir

The physical reservoirs listed earlier were classical. A register of interacting qubits is also a driven physical system with many readable quantities, and quantum reservoir computing uses it in place of the random network [35, 36]. The sample is written into one input qubit, and the register evolves for a time Δt under a fixed Hamiltonian. The expectation values of a set of observables form the feature vector, and only a linear readout of these values is fitted.

11.1 Qubits and the state vector

A qubit is a two-level quantum system [37], for instance a spin one half, with two basis states written $|0\rangle$ and $|1\rangle$. Its general state is a superposition

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle, \qquad |a_0|^2 + |a_1|^2 = 1, \tag{11.1}$$

with complex amplitudes a_0 and a_1. A measurement in the basis $\{|0\rangle, |1\rangle\}$ gives 0 with probability $|a_0|^2$ and 1 with probability $|a_1|^2$, the Born rule. A register of N qubits has the 2^N basis states $|z_1 z_2 \dots z_N\rangle$ with every $z_i \in \{0, 1\}$, and its state is a vector of 2^N complex amplitudes.

Measurable quantities are Hermitian operators, the observables. For one qubit the basic ones are the Pauli matrices

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{11.2}$$

in the basis $|0\rangle, |1\rangle$, and $\hat{X}_i$, $\hat{Y}_i$, $\hat{Z}_i$ denote these matrices acting on qubit i of a register, with the identity on all other qubits. The mean of an observable $\hat{P}$ over many repetitions of the same preparation and measurement is the expectation value $\langle \hat{P} \rangle = \langle \psi | \hat{P} | \psi \rangle$. For the state $|\psi(s)\rangle = \sqrt{1-s}\,|0\rangle + \sqrt{s}\,|1\rangle$ with $0 \le s \le 1$, which writes a number s into a qubit,

$$\langle \hat{Z} \rangle = (1 - s) - s = 1 - 2s, \qquad \langle \hat{X} \rangle = 2\sqrt{s(1-s)}, \qquad \langle \hat{Y} \rangle = 0. \tag{11.3}$$

The first is linear in s, and the second is nonlinear.

11.2 The density matrix

A state vector describes a register whose preparation is known completely. A reservoir discards a qubit at every step, and the remaining qubits alone are in general not described by any state vector. This case, and a mixture of states $|\psi_n\rangle$ with probabilities p_n, are covered by the density matrix

$$\hat{\rho} = \sum_n p_n \, |\psi_n\rangle\langle\psi_n|, \qquad \langle \hat{P} \rangle = \mathrm{Tr}(\hat{P}\hat{\rho}), \tag{11.4}$$

a Hermitian $2^N \times 2^N$ matrix with non-negative eigenvalues and unit trace, fixed by $4^N - 1$ real numbers. A pure state has $\hat{\rho} = |\psi\rangle\langle\psi|$. From here on $\hat{\rho}$ denotes a density matrix, and the spectral radius of the echo state network no longer appears.

For one qubit, the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ and the equal mixture of $|0\rangle$ and $|1\rangle$ have

$$\hat{\rho}_+ = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{mix}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Both give the outcomes 0 and 1 with equal probabilities. They differ in the off-diagonal elements, the coherences, which the observable $\hat{X}$ detects, since $\langle \hat{X} \rangle = \mathrm{Tr}(\hat{X}\hat{\rho})$ is 1 for $\hat{\rho}_+$ and 0 for $\hat{\rho}_{\mathrm{mix}}$.

11.3 The partial trace

The operation that describes the remaining qubits after one is discarded is the partial trace. For two qubits, with qubit 1 written first in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, the density matrix splits into four 2×2 blocks, one for each pair of states of qubit 1. The trace over qubit 1 adds the two diagonal blocks,

$$\hat{\rho} = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \quad \Longrightarrow \quad \mathrm{Tr}_1 \hat{\rho} = \rho_{00} + \rho_{11}. \tag{11.5}$$

The result is the density matrix of qubit 2 alone, and it gives every expectation value of an observable of qubit 2.

For the product state $|\psi(s)\rangle \otimes |0\rangle$ only the elements with qubit 2 in $|0\rangle$ are nonzero, $\rho_{00} = \mathrm{diag}(1-s, 0)$ and $\rho_{11} = \mathrm{diag}(s, 0)$, and $\mathrm{Tr}_1 \hat{\rho} = |0\rangle\langle 0|$. Discarding qubit 1 leaves qubit 2 in its pure state, and the number s is lost. For the entangled state $(|00\rangle + |11\rangle)/\sqrt{2}$,

$$\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \rho_{00} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_{11} = \frac{1}{2}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \mathrm{Tr}_1 \hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The two qubits together are in a pure state, whereas qubit 2 alone is in the mixture $\hat{\rho}_{\mathrm{mix}}$. The correlation with qubit 1 is lost when qubit 1 is discarded. In the reservoir the discarded qubit is in general entangled with the rest in this way.

11.4 Linear evolution

Between two inputs the register is closed and evolves by the Schrödinger equation with a Hamiltonian $\hat{H}$, a Hermitian operator. Over a time Δt a state vector is multiplied by the unitary operator $\hat{U} = e^{-i\hat{H}\Delta t}$, in units with $\hbar = 1$, and a density matrix becomes

$$\hat{\rho} \;\mapsto\; \hat{U}\hat{\rho}\,\hat{U}^{\dagger}. \tag{11.6}$$

This map is linear in $\hat{\rho}$ and preserves distances between states. The trace distance $\mathcal{D}(\hat{\rho}, \hat{\sigma}) = \frac{1}{2}\sum_m |\lambda_m|$, with λ_m the eigenvalues of $\hat{\rho} - \hat{\sigma}$, is the largest difference between the probabilities that the two states give to any measurement outcome. The matrix $\hat{U}(\hat{\rho} - \hat{\sigma})\hat{U}^{\dagger}$ has the same eigenvalues as $\hat{\rho} - \hat{\sigma}$, so $\mathcal{D}$ does not change under equation (11.6). A closed register therefore never forgets its start, the opposite of the echo state property.

Forgetting and nonlinearity, the two ingredients that a closed register lacks, have to come from the interface with the outside. Qubits 2 to N keep the memory, and the reset of the input qubit acts as its eraser. The partial trace discards whatever qubit 1 holds before the new input is written in its place. A partial trace can only shrink the trace distance between two states, whereas a unitary step keeps it. One reset removes only what sits on qubit 1. The couplings keep carrying information from the other qubits onto it, however, and with every reset a little more of the past is erased and old inputs fade. The nonlinearity comes from the way the input is written, as the expectation values (11.3) already show.

11.5 The spin network

The reservoir is a network of N qubits, six in the reservoir simulations, in which every pair is coupled [35, 38],

$$\hat{H} = \sum_{i<j} J_{ij}\, \hat{X}_i \hat{X}_j + \frac{1}{2}\sum_{i=1}^{N} (h + D_i)\hat{Z}_i. \tag{11.7}$$

Each coupling $\hat{X}_i\hat{X}_j$ flips the qubits i and j together, with a strength J_{ij} drawn once from the uniform distribution on $[-J_s/2, J_s/2]$. The scale J_s sets the unit of energy and of inverse time. The

transverse field h splits the two states of every qubit by the same energy. The random fields D_i, uniform on $[-D, D]$ with the disorder strength D, give each qubit its own splitting, so that a flip-flop $|\dots 0_i \dots 1_j \dots\rangle \to |\dots 1_i \dots 0_j \dots\rangle$ produced by $\hat{X}_i \hat{X}_j$ costs the energy $|D_i - D_j|$.

Two symmetries hold for all h and D. The matrix of $\hat{H}$ in the basis $|z_1 \dots z_N\rangle$ is real, so $\hat{H}$ is invariant under time reversal. Every coupling flips two spins, so $\hat{H}$ commutes with the parity $\hat{\Pi} = \hat{Z}_1 \hat{Z}_2 \cdots \hat{Z}_N$ and splits into an even and an odd sector of 2^{N-1} levels each.

Three parameter points represent the three regimes of this register. At the ergodic point $(h, D) = (10, 0)\, J_s$ no local quantity is conserved, and the state spreads over its whole parity sector. Local observables relax to the values of a thermal state at the same energy, which is called thermalisation. At the spin-glass point $(0.01, 0)\, J_s$ the field is too weak to compete with the couplings. The operators $\hat{X}_j$ are then nearly conserved, and the register stays frozen close to its initial configuration. At the localised point $(1, 100)\, J_s$ the random fields dominate, the $\hat{Z}_i$ are nearly conserved, and an excitation stays on the qubit where it was created.

11.6 One reservoir step

Figure 43 draws one input step on the register, qubit 1 receiving the input and qubits 2 to N carrying the memory. The register enters the step in the state $\hat{\rho}_{k-1}$ left by the previous one. (i) Qubit 1 is discarded, which leaves the memory qubits in the reduced state $\mathrm{Tr}_1 \hat{\rho}_{k-1}$. Qubit 1 is then prepared anew in the state $|\psi(s_k)\rangle = \hat{R}_y(\theta_k)|0\rangle$, with the rotation $\hat{R}_y(\theta) \coloneqq e^{-i\theta \hat{Y}/2}$ about the y axis and the angle $\theta_k = 2 \arcsin \sqrt{s_k}$. Since $\hat{R}_y(\theta)|0\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle$, this gives the amplitudes $|\psi(s_k)\rangle = \sqrt{1 - s_k}\,|0\rangle + \sqrt{s_k}\,|1\rangle$. (ii) The whole register evolves under the fixed Hamiltonian for the time Δt,

$$\hat{\rho}_k = \mathcal{T}_{s_k}(\hat{\rho}_{k-1}) \coloneqq \hat{U}\Big(|\psi(s_k)\rangle\langle\psi(s_k)| \otimes \mathrm{Tr}_1\, \hat{\rho}_{k-1}\Big)\hat{U}^\dagger, \qquad \hat{U} = e^{-i\hat{H}\Delta t}, \tag{11.8}$$

so that after k inputs the register is in the state $\hat{\rho}_k = \mathcal{T}_{s_k} \circ \cdots \circ \mathcal{T}_{s_1}(\hat{\rho}_0)$. (iii) The features are the expectation values of Pauli strings, products of Pauli matrices on one or two qubits,

$$x_{k,a} = \mathrm{Tr}(\hat{P}_a\, \hat{\rho}_k), \qquad \hat{P}_a \in \{\hat{X}_i,\ \hat{Y}_i,\ \hat{Z}_i,\ \hat{X}_i\hat{X}_j,\ \hat{Y}_i\hat{Y}_j,\ \hat{Z}_i\hat{Z}_j\}, \tag{11.9}$$

given by the Born rule and computed without sampling noise from $\hat{\rho}_k$ in the simulation. There are $M = 3N + 3N(N-1)/2$ of them, far fewer than the $4^N - 1$ numbers that fix $\hat{\rho}_k$. (iv) The linear readout $\tilde{\mathbf{y}}_k = W_{\mathrm{out}}^\top \mathbf{x}_k + \mathbf{b}$ of equation (2.1) is the only trained part, fitted by ridge regression, and $\hat{\rho}_k$ enters the next step.

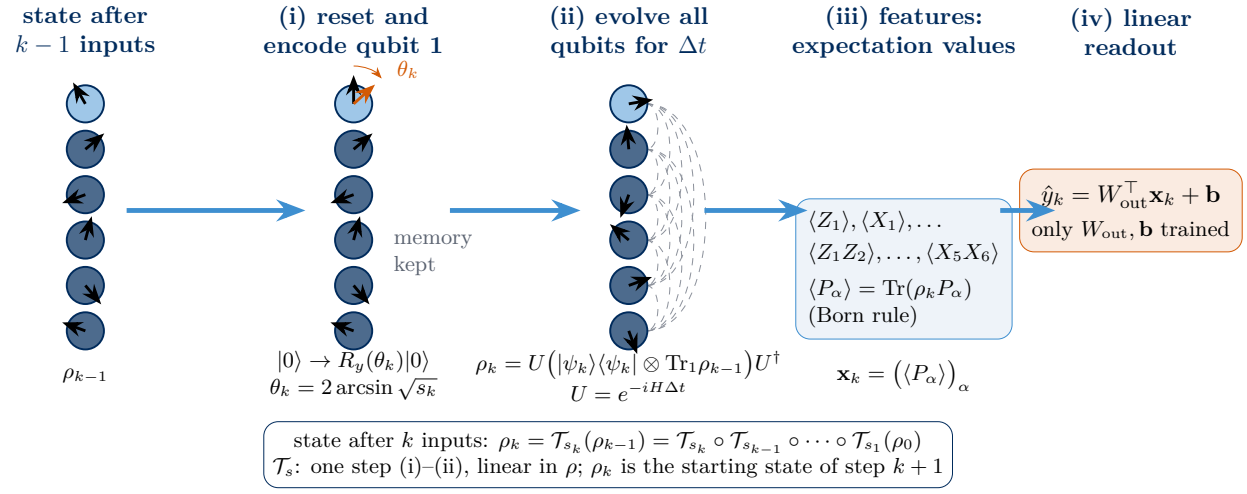


Figure 43: One step of the quantum reservoir, starting from the state $\hat{\rho}_{k-1}$ after $k-1$ inputs. The step consists of (i) the reset of qubit 1 and its encoding by the angle θ_k and (ii) the evolution of all qubits under $\hat{H}$ for the time Δt. It ends with (iii) the Born-rule expectation values $\mathbf{x}_k$ and (iv) the linear readout. Qubits 2 to N keep the memory and are never reset.

The step (11.8) is linear in $\hat{\rho}_{k-1}$, since discarding a qubit, attaching a fixed state and conjugating with $\hat{U}$ are all linear operations on density matrices. The nonlinearity in s_k enters only through the encoding, whose amplitudes $\sqrt{s_k}$ and $\sqrt{1-s_k}$ appear in pairs. Writing $\hat{U}_0$ and $\hat{U}_1$ for the parts of $\hat{U}$ that act on the states with qubit 1 in $|0\rangle$ and in $|1\rangle$, and $\hat{\sigma} = \mathrm{Tr}_1 \hat{\rho}_{k-1}$,

$$\begin{aligned} \mathcal{T}_s(\hat{\rho}_{k-1}) = (1-s)\,\hat{U}_0\hat{\sigma}\hat{U}_0^\dagger + s\,\hat{U}_1\hat{\sigma}\hat{U}_1^\dagger \\ + \sqrt{s(1-s)}\,(\hat{U}_0\hat{\sigma}\hat{U}_1^\dagger + \hat{U}_1\hat{\sigma}\hat{U}_0^\dagger). \end{aligned} \tag{11.10}$$

After one step every feature is a combination of the three functions $1-s$, s and $\sqrt{s(1-s)}$ of the input. A fit with these three functions reproduces all computed features up to rounding errors. Over many steps such functions of $s_k, s_{k-1}, \ldots$ multiply, and the features become products of functions of past inputs, the structure of the Volterra series (5.3).

Two copies differ by $\hat{\Delta} = \hat{\rho}_{k-1} - \hat{\rho}'_{k-1}$, and the step maps $\hat{\Delta}$ to $\hat{U}(|\psi\rangle\langle\psi| \otimes \mathrm{Tr}_1\hat{\Delta})\hat{U}^\dagger$. Its trace distance equals that of $\mathrm{Tr}_1\hat{\Delta}$, which is at most that of $\hat{\Delta}$. The step therefore forgets, and the discarded qubit is the only place where the register loses information. Table 7 sets the parts of this reservoir next to those of the echo state network.

	echo state network	quantum reservoir
state	$\mathbf{r}_k \in \mathbb{R}^N$	density matrix $\hat{\rho}_k$, $4^N - 1$ real numbers
input enters	$\mathbf{w}_{\mathrm{in}} s_k$ added to every unit	qubit 1 prepared in $\|\psi(s_k)\rangle$
fixed dynamics	tanh of $W\mathbf{r}_{k-1} + \mathbf{w}_{\mathrm{in}} s_k$	$\hat{U} = e^{-i\hat{H}\Delta t}$, linear in $\hat{\rho}$
nonlinearity	curvature of tanh	amplitudes $\sqrt{s}$, $\sqrt{1-s}$ of the encoding
forgetting	contraction by the leak and by W	discarding qubit 1
control	spectral radius, σ_{in}, α	field h, disorder D, time Δt
features	the state itself	M Pauli expectation values

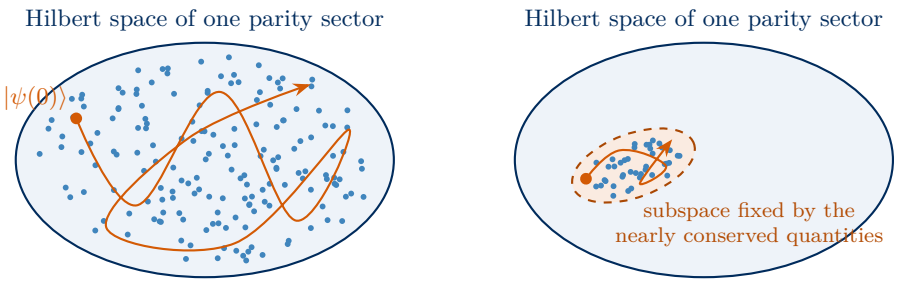
Table 7: The echo state network and the quantum reservoir, part by part.

11.7 Quantum chaos and the level-spacing ratio

Whether the discarded qubit removes the memory of the start depends on how the register moves through its Hilbert space between two inputs. In a chaotic quantum system an initial state explores, at late times, the whole part of the Hilbert space allowed by the symmetries, here one parity sector (Figure 44, left). Information written into one qubit spreads into correlations among all qubits, a process called scrambling. The expectation values of operators on that qubit, or on any few qubits, then lose it. In a near-integrable or localised system nearly conserved quantities keep their initial values, together with the information they carry, and the state stays in the small subspace they fix (right).

An operator $\hat{\mathcal{O}}$ is conserved when $[\hat{H}, \hat{\mathcal{O}}] = 0$, since $\frac{\mathrm{d}}{\mathrm{d}t}\langle\hat{\mathcal{O}}\rangle = i\langle[\hat{H}, \hat{\mathcal{O}}]\rangle$ with $\hbar = 1$. A chaotic system conserves only the energy and a few symmetries, here the parity, while an integrable system has a number of independent conserved quantities that grows with N. For a reservoir, scrambling is the mechanism that erases the initial state and mixes every input into all features.

The spin network has both behaviours. Without field and disorder every $\hat{X}_j$ commutes with every coupling $\hat{X}_i\hat{X}_l$, so the N operators $\hat{X}_j$ are conserved. A small field breaks this weakly, since $[\hat{H}, \hat{X}_j] = i(h + D_j)\hat{Y}_j$. At strong disorder every flip-flop costs an energy much larger than the couplings can supply, and each $\hat{Z}_i$ is nearly conserved. This localised regime of the small register is a finite-size relative of many-body localisation [42]. Figure 45 follows an imprint on qubit 1 under $\hat{H}$ alone at the three parameter points. At the ergodic point both the $\hat{Z}$ and the $\hat{X}$ imprint are



chaotic
the state spreads over the whole sector

near-integrable or localised
the state stays in a small subspace

Figure 44: The time evolution of one state, sketched as points visited at late times (blue) and a path (orange) from the initial state $|\psi(0)\rangle$. Left, chaotic dynamics. At late times the state has explored the whole Hilbert space allowed by the symmetries, here one parity sector. The memory of $|\psi(0)\rangle$ is spread over correlations of all qubits. Right, near-integrable or localised dynamics. The nearly conserved quantities keep their initial values, and the state stays in the small subspace they fix.

gone within one reservoir step. The value of $\langle \hat{Z}_1 \rangle$ then fluctuates around a positive constant, the thermal value fixed by the energy of the initial state, which an ergodic register shares among all its qubits. At the spin-glass point $\langle \hat{X}_1 \rangle$ stays at its initial value, held by the nearly conserved $\hat{X}_1$, and at the localised point $\langle \hat{Z}_1 \rangle$ does. Each non-ergodic regime keeps one kind of information, and only the ergodic regime scrambles both.

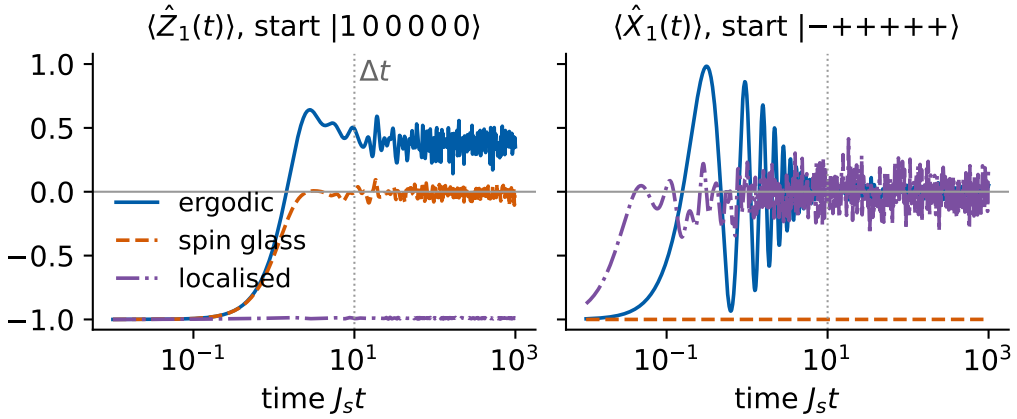


Figure 45: An imprint on qubit 1 under the Hamiltonian alone, without inputs or resets, at an ergodic, a spin-glass and a localised parameter point. Left, $\langle \hat{Z}_1(t) \rangle$ after a flip of qubit 1 in the $\hat{Z}$ basis. Right, $\langle \hat{X}_1(t) \rangle$ after a flip in the $\hat{X}$ basis, $\hat{X}|\pm\rangle = \pm|\pm\rangle$. Dotted, the time Δt between two inputs of the reservoir.

The energy spectrum distinguishes these two behaviours. The energies are the eigenvalues of the Hamiltonian, $\hat{H}|E_n\rangle = E_n|E_n\rangle$, ordered as $E_1 \le E_2 \le \dots$ within one parity sector, with the spacings $\delta_n := E_{n+1} - E_n$. We follow two levels as a parameter of $\hat{H}$ is varied (Figure 46). When the two eigenstates belong to different symmetry sectors, $\hat{H}$ has no matrix element between them, and their energies pass through each other in a crossing. When they belong to the same sector, any coupling V between the two approaching states mixes them. The two-level problem with the energies $\varepsilon_1, \varepsilon_2$ and the coupling V has the eigenvalues $\frac{1}{2}(\varepsilon_1 + \varepsilon_2) \pm \sqrt{\frac{1}{4}(\varepsilon_1 - \varepsilon_2)^2 + |V|^2}$, which stay at least $2|V|$ apart. The levels repel each other in an avoided crossing, the no-crossing rule [39].

In a near-integrable system the nearly conserved quantities split $\hat{H}$ into many nearly independent blocks, and the levels of different blocks cross freely. The full spectrum is then a superposition of many unrelated sequences, its spacings follow Poisson statistics, and small gaps are the most frequent. In a chaotic system the states within each parity sector are all coupled, approaching levels repel, and small gaps become rare, which is called level repulsion. The spacings follow the Wigner–Dyson statistics of random matrices. For a real Hamiltonian such as (11.7) the reference is the Gaussian orthogonal ensemble (GOE) of real symmetric random matrices, whose spacing density vanishes linearly at zero

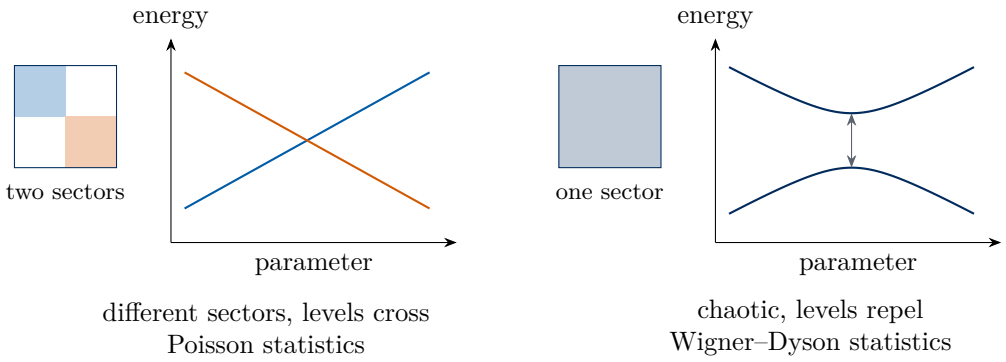


Figure 46: Two energy levels against a parameter of the Hamiltonian, sketched, with the block structure of $\hat{H}$ on the left of each panel. Left, the two states belong to different symmetry sectors, $\hat{H}$ has no matrix element between them, and their energies cross. Right, the two states belong to the same sector, any coupling mixes them, and the levels repel and keep a gap (grey arrow).

spacing.

The local density of levels varies across the spectrum, so raw spacings of different parts cannot be compared directly. The ratio of two neighbouring spacings (Figure 47)

$$r_n \coloneqq \frac{\min(\delta_n, \delta_{n-1})}{\max(\delta_n, \delta_{n-1})} \in [0, 1] \tag{11.11}$$

does not depend on the local density [40]. The ratio is small when one of two neighbouring gaps is much smaller than the other, so its mean measures how often small gaps occur. For Poisson statistics $P(r) = 2/(1+r)^2$, largest at $r = 0$, with the mean $\langle r \rangle = 2\ln 2 - 1 \approx 0.386$. For the GOE the distribution is close to $P(r) = \frac{27}{4}(r + r^2)/(1 + r + r^2)^{5/2}$, which vanishes at $r = 0$, with the mean $\langle r \rangle \approx 0.53$ [41]. For the spin network $\langle r \rangle$ is averaged over the central half of the spectrum and over draws of J_{ij} and D_i.

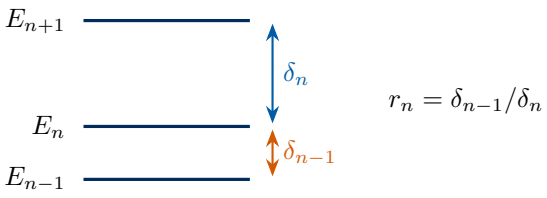


Figure 47: The gap ratio on three neighbouring levels. Here $\delta_{n-1} < \delta_n$, and r_n is about one half.

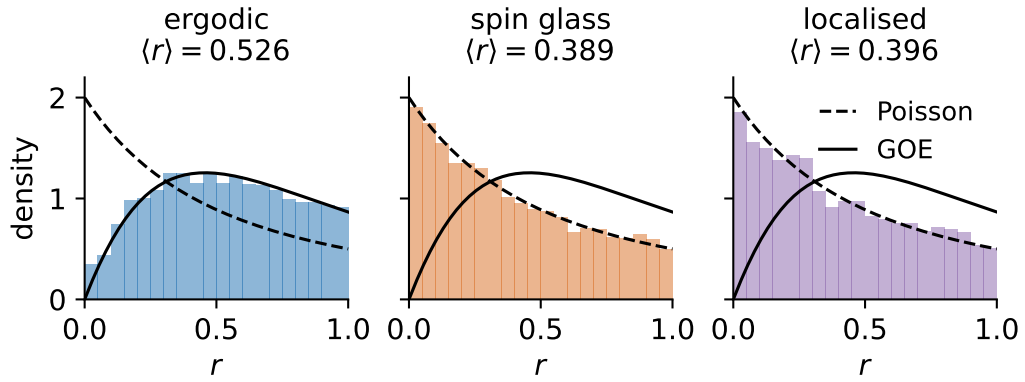


Figure 48: Histograms of the gap ratio r_n for ten qubits at the ergodic, spin-glass and localised points. Dashed, the Poisson distribution. Solid, the GOE distribution.

Figure 48 shows both cases for ten qubits. At the ergodic point the histogram follows the GOE curve and drops towards $r = 0$, and its mean reaches the GOE value. At the spin-glass and localised

points it follows the Poisson curve and is largest at $r = 0$, the mark of the nearly conserved $\hat{X}_j$ and $\hat{Z}_i$.

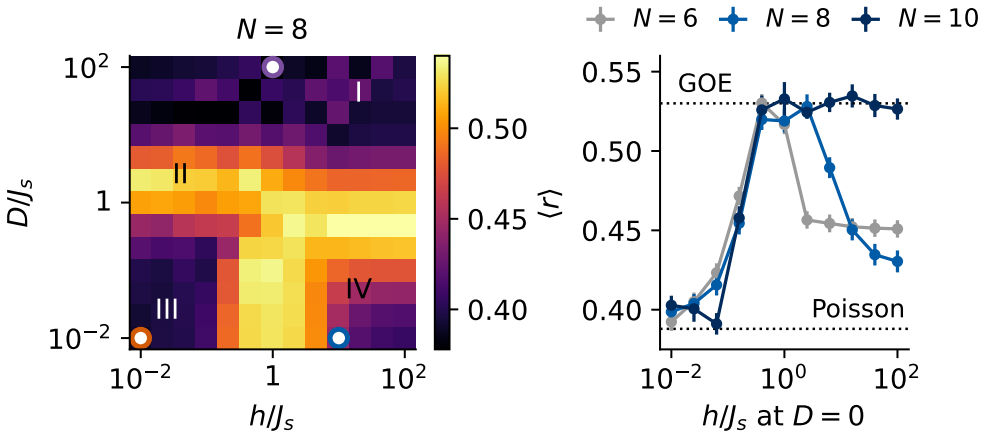


Figure 49: The mean gap ratio $\langle r \rangle$ of the spin network. Left, over the field h and the disorder strength D for eight qubits. Circles mark the three parameter points, with $D = 0$ drawn on the lowest row, and the regions are I localised, II and IV ergodic, III spin glass. Right, against h at $D = 0$ for six, eight and ten qubits, with the Poisson and GOE values (dotted).

Computed over the plane of field and disorder, $\langle r \rangle$ is a map of quantum chaos in the spin network (Figure 49), in agreement with [38]. Strong disorder localises the register. The random fields make every flip-flop too costly, each $\hat{Z}_i$ is nearly conserved, and the statistics is Poisson. With a weak field and weak disorder the couplings dominate. The Hamiltonian $\hat{H}$ is then close to a classical Ising model in the $\hat{X}$ basis with couplings of random sign, a spin glass. Its nearly conserved $\hat{X}_j$ again give Poisson statistics. Without disorder a flip-flop costs no energy, so under a strong field the couplings move flips freely between the qubits. In this case, and for disorder comparable to the couplings, nothing local is conserved, the levels repel and $\langle r \rangle$ approaches the GOE value. The register of the reservoir is small, and at six and eight qubits the ergodic point lies only partway between the Poisson and the GOE value, a finite-size effect. At ten qubits it reaches the GOE value.

The map predicts the behaviour of the reservoir before any training. A chaotic register scrambles every imprint and forgets its initial state, whereas a near-integrable register keeps part of the information about its start in the nearly conserved quantities.

11.8 Forgetting in three regimes

Two copies of the register start from different random density matrices and receive the same random inputs, and we follow their trace distance $\mathcal{D}_k$ (Figure 50). At the ergodic point it decays exponentially and reaches the level of rounding errors within a few hundred steps. At the spin-glass and localised points it stays large over the whole run.

The ergodic point, with $\langle r \rangle$ near the GOE value, scrambles the difference between the copies and forgets it. The two points with Poisson statistics keep it in their nearly conserved quantities. The reset of qubit 1 leaves $\hat{X}_2, \ldots, \hat{X}_N$ untouched, so without field and disorder the differences of $\langle \hat{X}_j \rangle$ between the copies survive every step. The small field of the spin-glass point removes them only slowly. At the localised point the other qubits keep their nearly conserved $\langle \hat{Z}_j \rangle$, and qubit 1, whose state is replaced at every step, exchanges little with them.

Without the echo state property a readout works only for the start used in training. We apply a readout fitted on a run from the maximally mixed state to a run of the same reservoir with the same inputs started from a random pure state. The readout of the spin-glass reservoir, whose recall capacity on the training start even exceeds that of the ergodic one, loses all of it on the second run. Its recall was carried by the memory of its start, which the inputs never overwrite. The readout of the ergodic reservoir keeps its capacity unchanged.

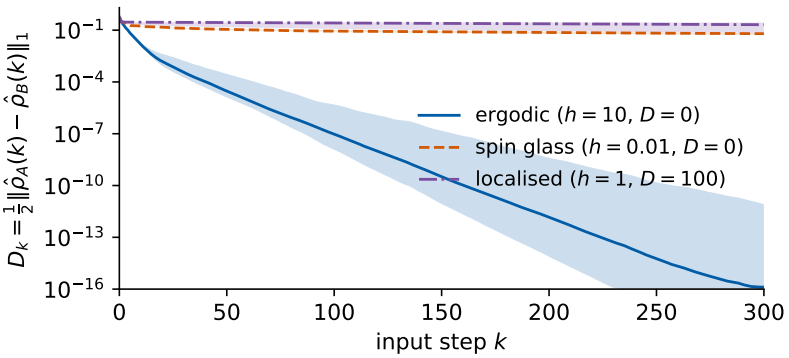


Figure 50: The trace distance $\mathcal{D}_k$ between two copies of the six-qubit register against the input step k. The line is the median, and the band spans the 16th to the 84th percentile over ten draws of the couplings. The axis label writes it as $\frac{1}{2}\|\hat{\rho}_A(k) - \hat{\rho}_B(k)\|_1$, with $\hat{\rho}_A(k)$ and $\hat{\rho}_B(k)$ the density matrices of the two copies and $\|\cdot\|_1$ the sum of the moduli of the eigenvalues.

11.9 Memory of the quantum reservoir

The recall capacities of equation (6.1) measure how well the reservoir stores past inputs (Figure 51). The localised reservoir has almost no memory, since the input stays on qubit 1 and is erased at the next reset. The ergodic reservoir remembers the last few inputs. The largest memory appears at a field near the border between the spin-glass and the ergodic region, where $\langle r \rangle$ rises from the Poisson towards the GOE value. An echo state network with as many units as the quantum reservoir has features stores about twice as much as the best quantum setting.

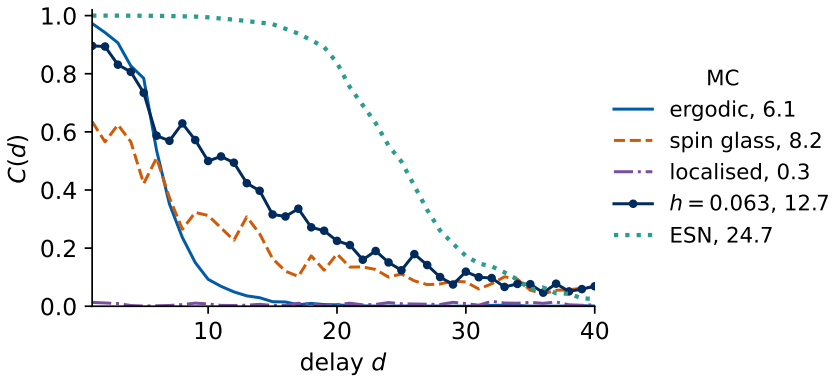


Figure 51: Recall capacity $C(d)$ against the delay d for the six-qubit reservoir in the three regimes and at the field of largest memory capacity, and for an echo state network with 63 units. The legend gives the memory capacities, and the curves are means over three draws.

11.10 Forecasting the laser with six qubits

The quantum reservoir forecasts the laser record with the protocol of the echo state network. We choose the field and the time step on the validation block and use one draw of the couplings. The reservoir reaches $C_{\text{tot}} = 24.4$, close to the 26.5 of an echo state network with as many units as the quantum reservoir has features and well above AR and persistence (Figure 52). One step ahead the quantum reservoir is the best of these models. These results come from one draw of the couplings and a small grid of settings.

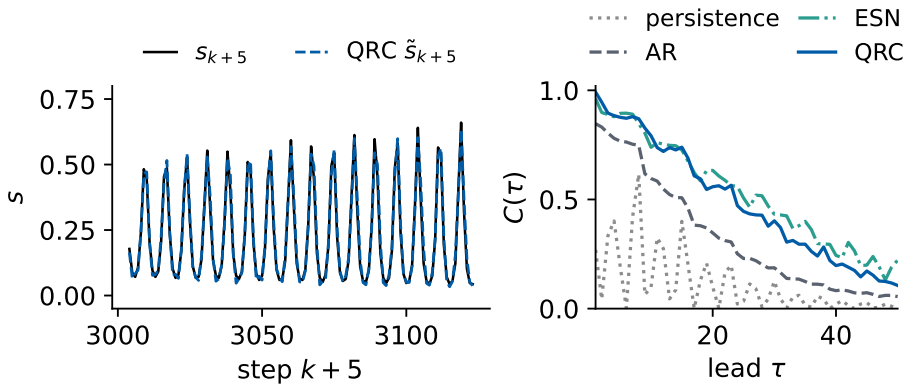


Figure 52: The six-qubit reservoir on the laser. Left, the target s_{k+5} and the forecast over the first 120 test origins. Right, $C(\tau)$ against the horizon τ, labelled lead in the figure, for persistence, AR, the echo state network with 63 units and the quantum reservoir.

11.11 Six qubits and a classical reservoir

A register of six qubits passes the reservoir tests in its ergodic regime and fails them in the spin-glass and localised regimes. The level statistics predict this outcome without any training. At this size it stores fewer past inputs than an echo state network with the same number of features and forecasts the laser about as well. No advantage over such a classical reservoir appears in these data. In an experiment every expectation value has to be estimated from many repeated runs. A projective measurement destroys the memory stored in the register, and the register must then be fed the input history again or measured only weakly after every step [43, 44]. Larger registers, decoherence and the use of dissipation as a resource are open problems of current research [45].

Each of the three reservoirs forgets by its own mechanism. The echo state network contracts its state with a random matrix, the cellular automaton overwrites its input cells and the qubit register discards its input qubit.

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